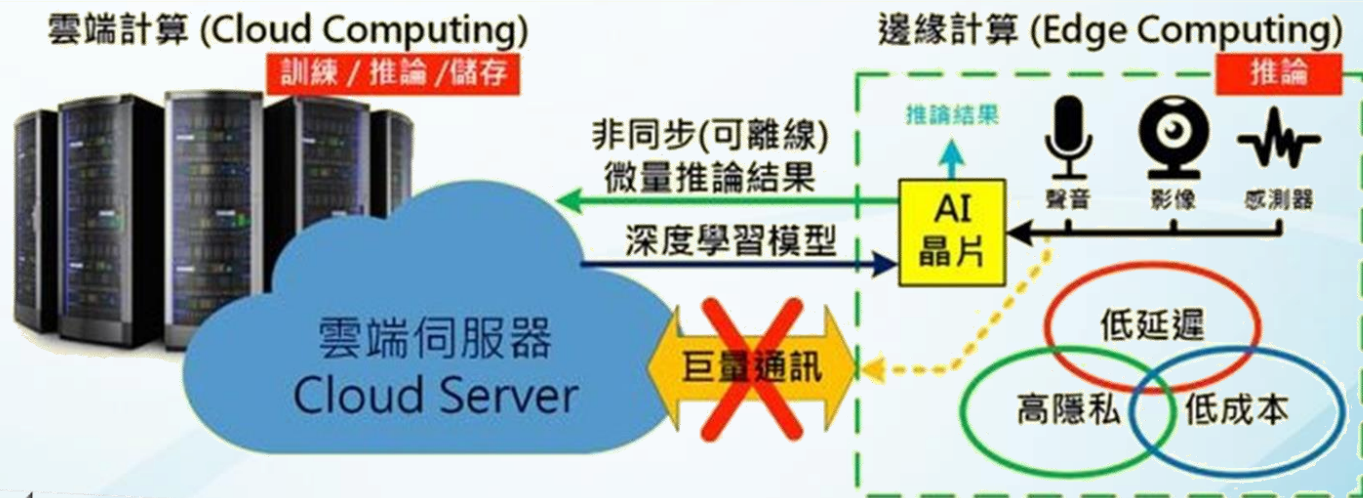
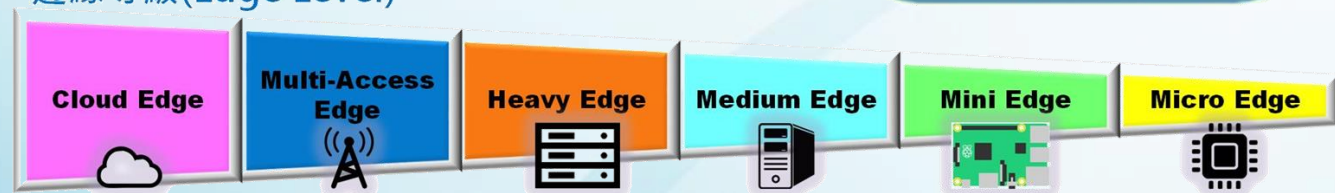


邊緣人工智慧實務 (EE5354701)



Edge AI

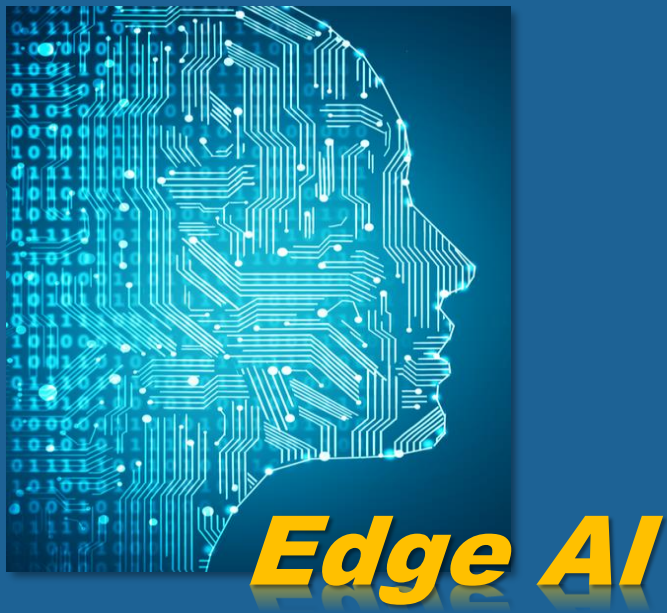
邊緣等級(Edge Level)



9 TinyML 案例實作 — 動作辨識

電機工程系 人工智慧應用產碩專班 許哲豪 兼任助理教授

簡報大綱

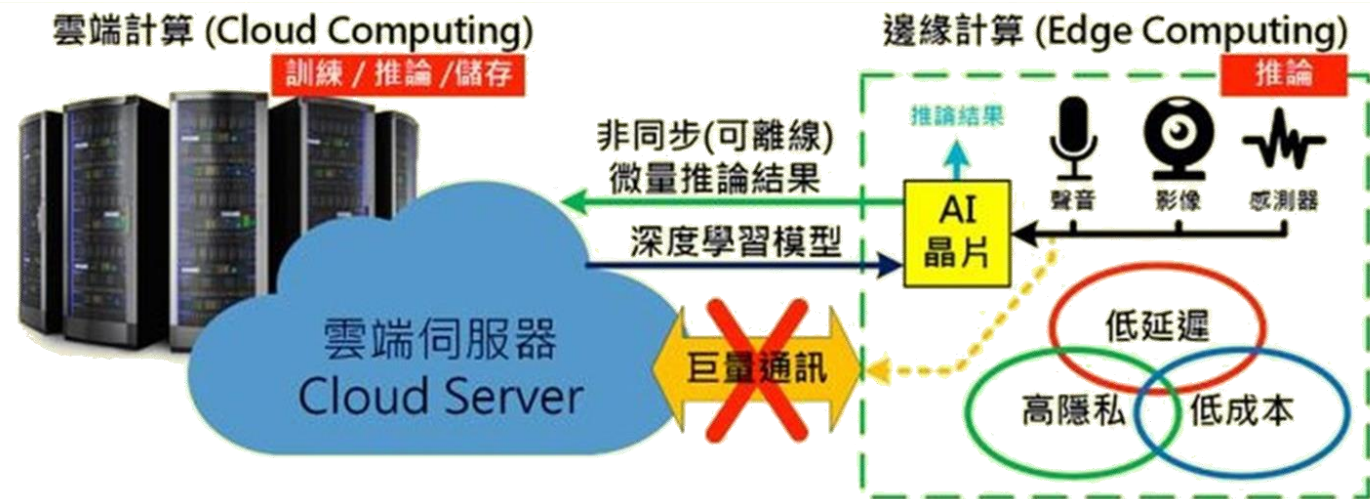


- 9.1. 運動感測器原理
- 9.2. 運動手勢辨識
- 9.3. 異常振動偵測

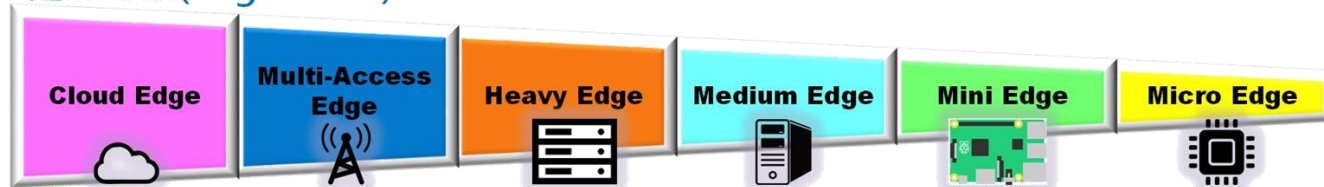
https://hackmd.io/@OmniXRI-Jack/NTUST_EdgeAI_2026



Edge AI



邊緣等級(Edge Level)



9.1. 運動感測器原理

何謂運動感測器 (Motion Sensor)



The image illustrates the components and concepts of motion sensors. On the left is a white Wii remote. In the center is a 3D coordinate system diagram with axes labeled Up, Down, Back, Forward, Left, and Right. Curved arrows indicate rotational degrees of freedom: Roll (green), Pitch (blue), and Yaw (orange). Below the diagram are three circuit boards: a blue MPU9250, a red LSM9DS1 Breakout, and a blue GY-85.

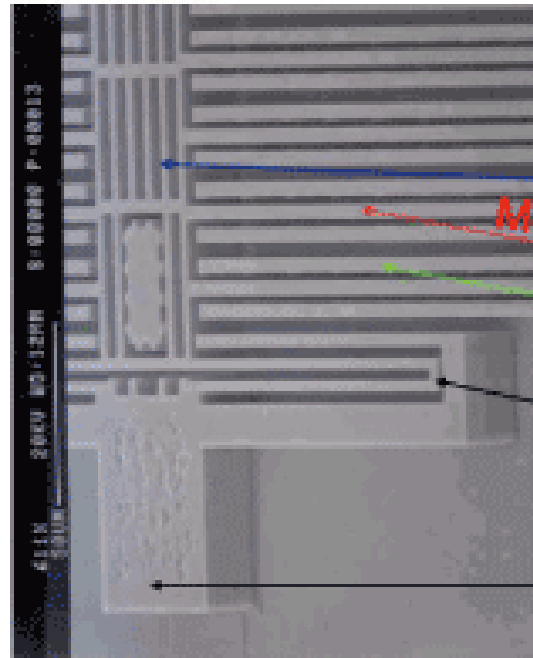
運動感測器 (Motion Sensor)

- 加速度計 (重力感測)
Accelerometer (G-Sensor)
- 陀螺儀 (旋轉量感測)
Gyroscope(Gyro-Sensor)
- 地磁感測計 (方向感測)
Magnetic(M-Sensor)
- 自由度(Degree of Freedom)
3 DoF, 6 DoF, 9 DoF
- 慣性量測單元 (IMU)
Inertial Measurement Unit

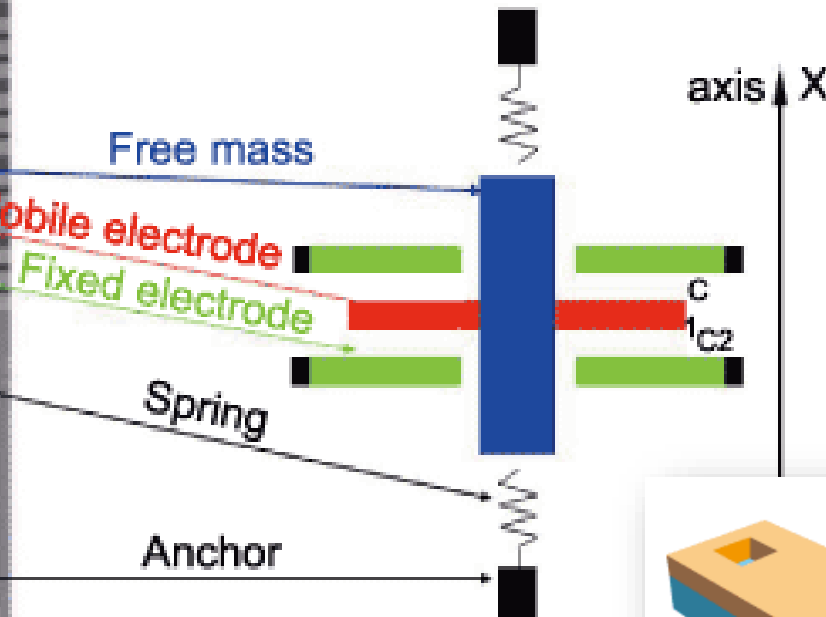
資料來源：<https://omnixri.blogspot.com/2022/04/20220408.html>

加速度計 (G Sensor)

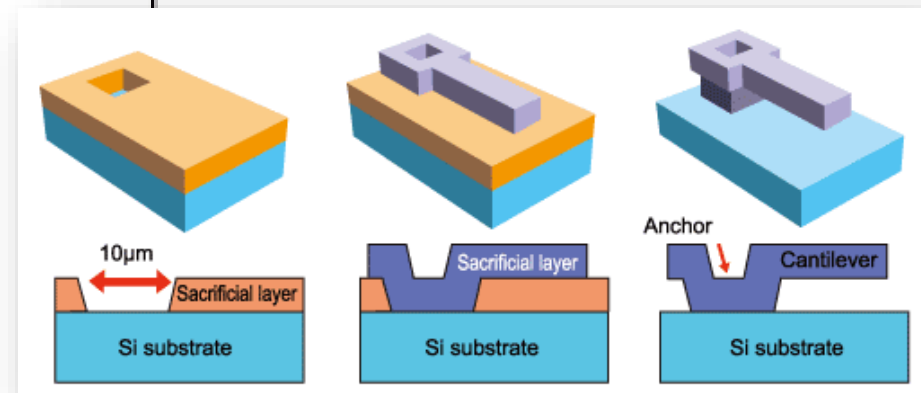
Silicon Mechanical Structure



MEMS Model

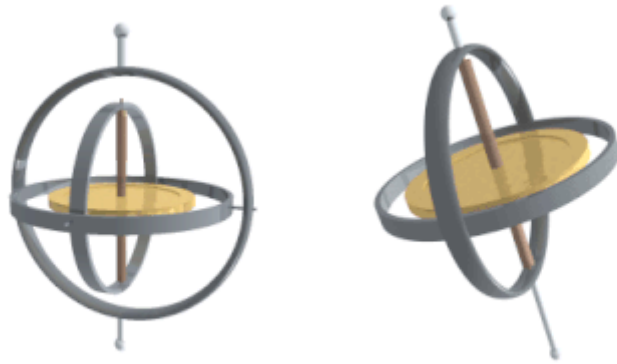


微機電MEMS 製程示意圖



資料來源：<https://www.ctimes.com.tw/DispArt/tw/%E7%BE%A9%E6%B3%95%E5%8D%8A%E5%B0%8E%E9%AB%94/MEMS/ST::%E5%8D%8A%E5%B0%8E%E9%AB%94/0711220000AW.shtml>

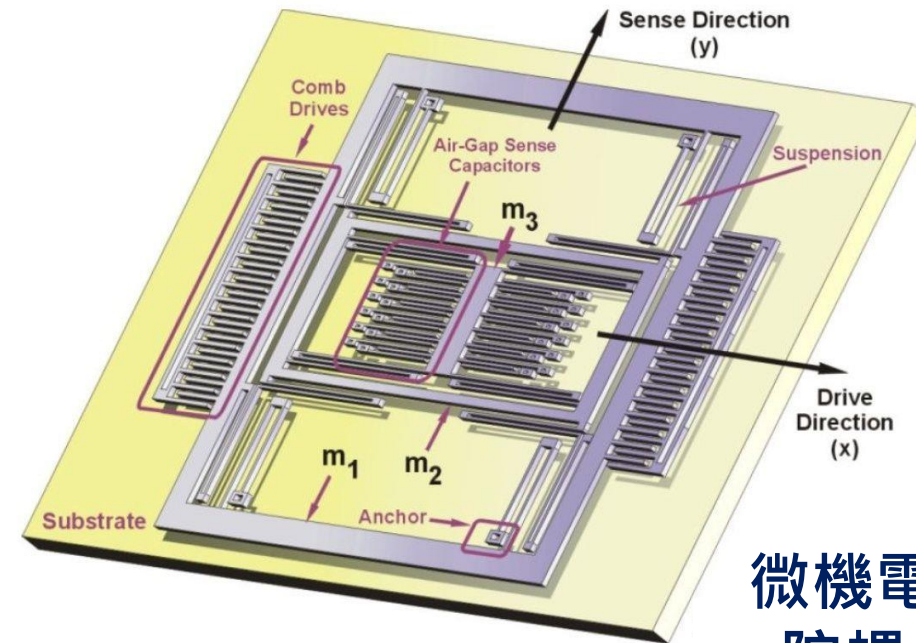
陀螺儀 (Gyro Sensor)



機械式陀螺儀



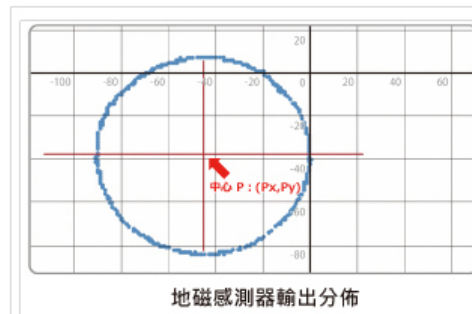
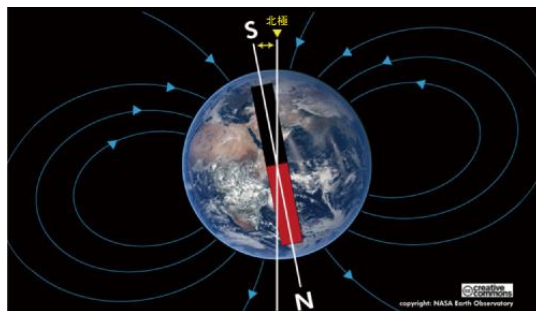
科里奧利力



微機電式
陀螺儀

資料來源：<https://zh.wikipedia.org/wiki/%E9%99%80%E8%9E%BA%E5%84%80>

地磁感測器 (Geomagnetic Sensor)



角度

角度 = $\tan^{-1} \frac{Y - P_y}{X - P_x}$

磁北角的計算公式

檢測方法	霍爾(Hall)	磁阻(MR)	磁抗(MI)
構成	基於磁場變化 電壓隨霍爾效應發生變化	基於磁場變化 MR元件的電阻值發生變化	基於磁場變化 利用脈衝電流檢測線圈電壓
抗雜訊 (靈敏度)	X	△	◎
消耗電流	X	△	◎
回應速度	X	△	◎

資料來源：https://www.rohm.com.tw/electronics-basics/sensors/sensor_what2

運動感測常見應用



人機互動
(Wii / PS Move 搖桿)



競速分析 / 駕駛習慣
(打龜號 / 車涯 04事件)



慣性導航 / 室內定位
(GPS無法看見天空)



異常振動
(馬達 / 輸送帶)

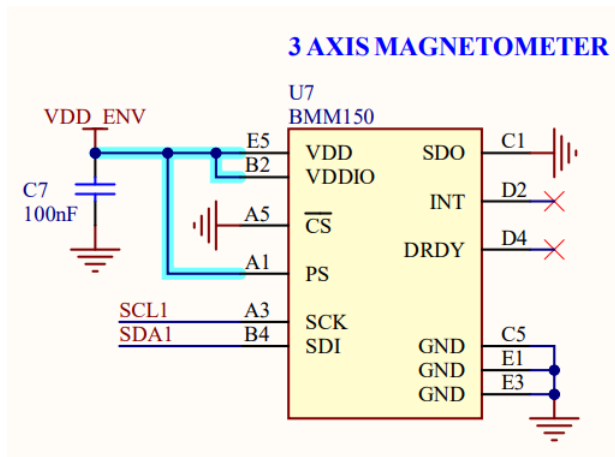


自動平衡
(平衡車、無人機、人形機器人)

Arduino Nano 33 BLE Sense Rev.2

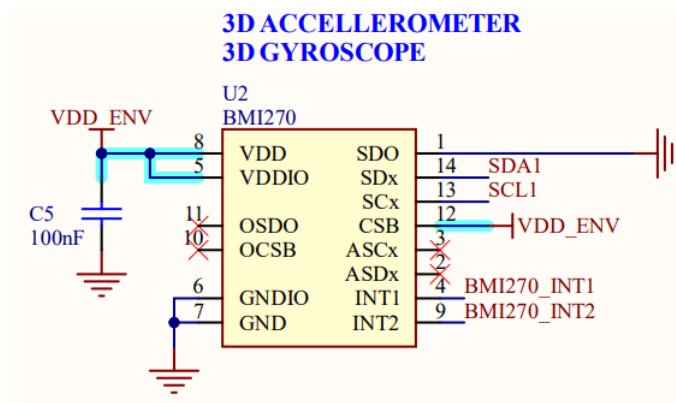
BMM150
Magnetometer 3 axis IMU

VDD	P0.22	
VDDIO	P0.22	
SDI	P0.14	(D) SDA1
SCK	P0.15	(D) SCL1
PS	P0.22	
SDO	GND	
CS	GND	



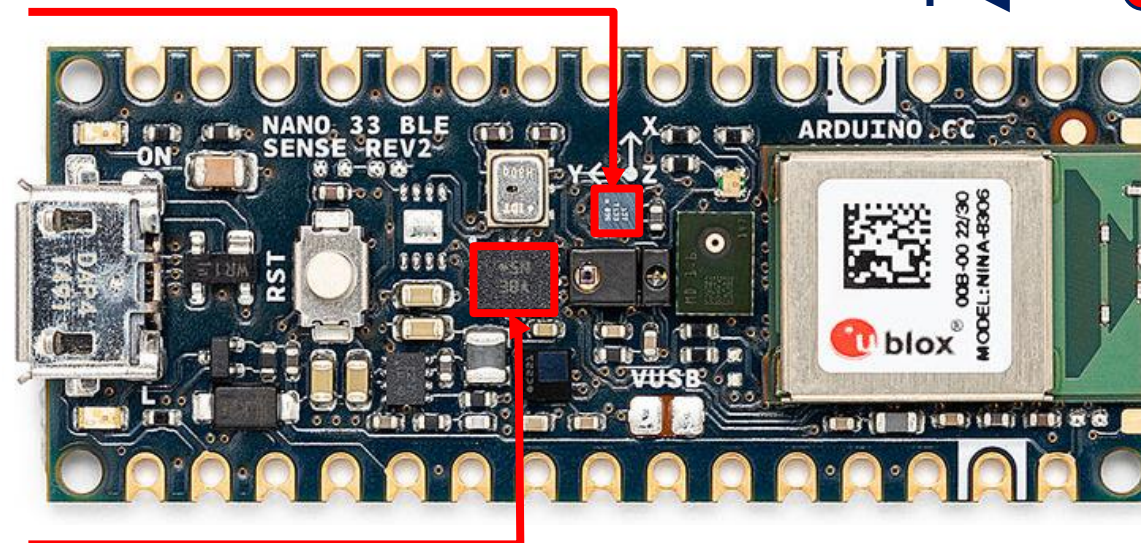
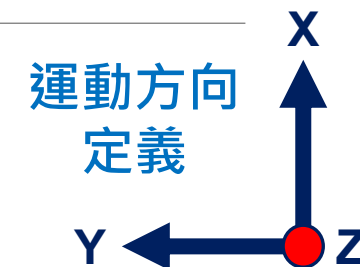
BMI270
Inertial Motion Unit 6 axis

P0.22	VDD	
P0.22	VDDIO	
SDA1 (D)	P0.14	SDx
SCL1 (D)	P0.15	SCx
P0.22	CSB	
P0.11	INT1	
P0.20	INT2	
GND	SDO	



BMM150 3軸地磁計

$\pm 1300\mu\text{T}$ (x, y-axis),
 $\pm 2500\mu\text{T}$ (z-axis) @16bit
ODR 2Hz ~ 30 Hz



BMI270

3軸加速度計

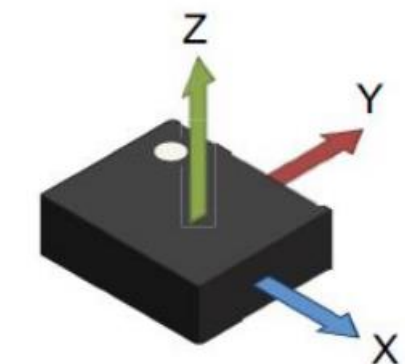
$\pm 2g/\pm 4g/\pm 8g/\pm 16g$ @16bit
ODR 0.78Hz ~ 1.6 kHz

3軸陀螺儀

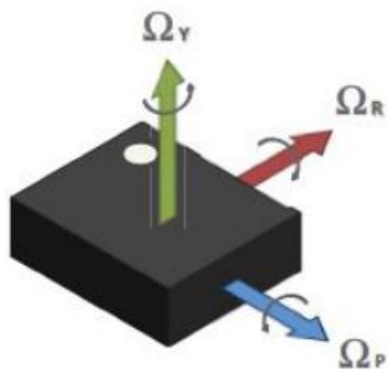
$\pm 125/\pm 250/\pm 500/\pm 1K/\pm 2K\text{dps}$ @16bit
ODR 25 Hz ~ 6.4 kHz

資料來源：<https://docs.arduino.cc/hardware/nano-33-ble-sense-rev2/>

Seeed Xiao nRF52840 Sense



Direction of detectable acceleration (top view)



Direction of detectable angular rate (top view)



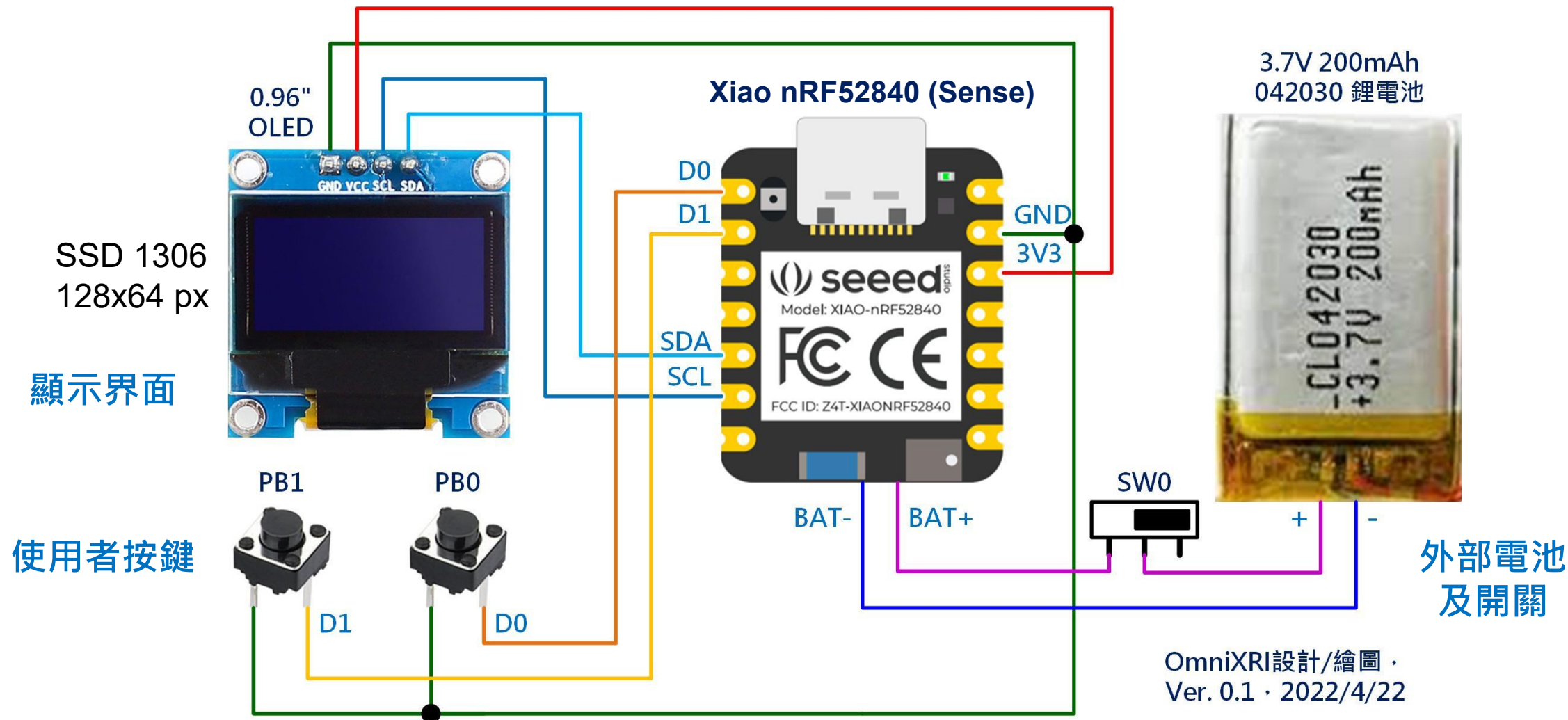
LSM6DS3TR-C (IMU)

3軸加速度計 $\pm 2g/\pm 4g/\pm 8g/\pm 16g$ @16bit
ODR 12.5Hz ~ 1.66KHz

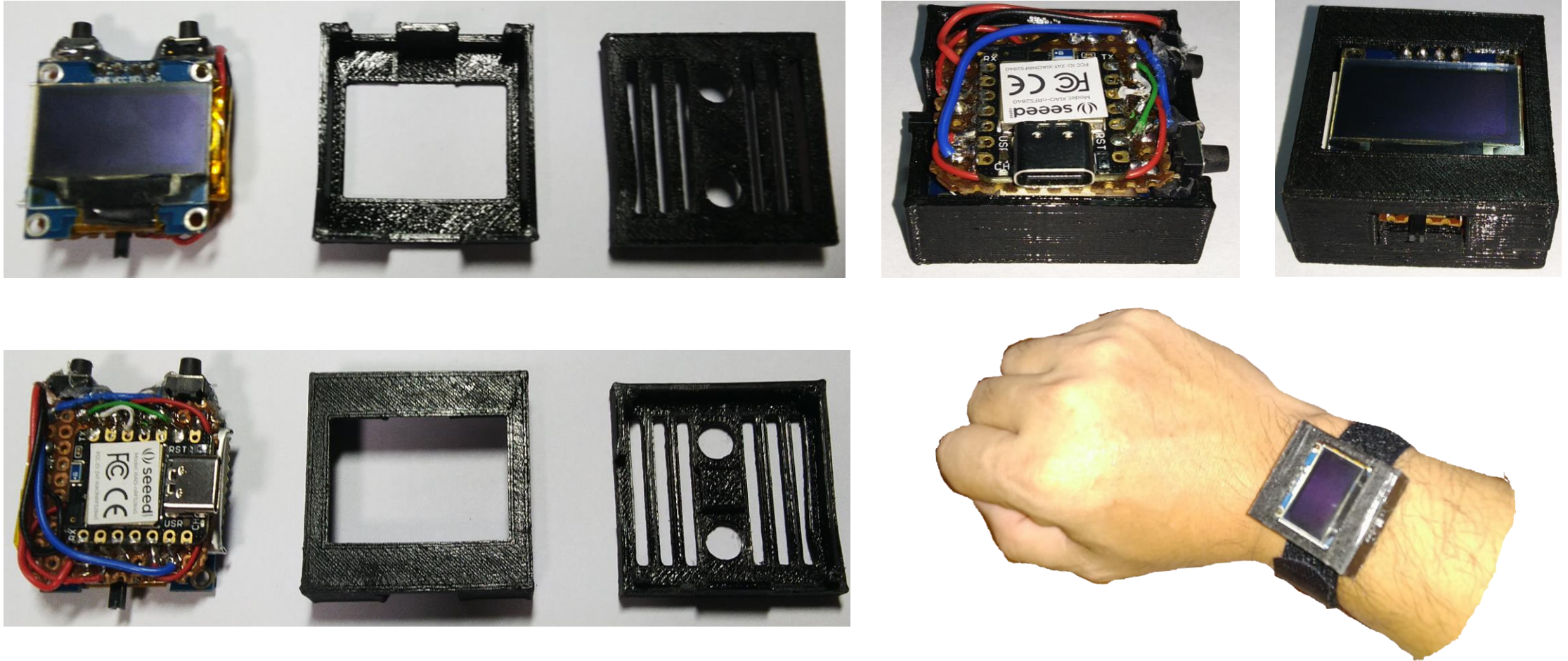
3軸陀螺儀 $\pm 125/\pm 250/\pm 500/\pm 1K/\pm 2Kdps$ @16bit
ODR 12.5Hz ~ 1.66KHz

溫度感測器 $-40 \sim +85\text{ }^{\circ}\text{C}$ @12bit
ODR 52Hz

穿戴式智慧人工智慧裝置 – 參考電路



穿戴式智慧人工智慧裝置 – 參考外形



資料來源：https://youtu.be/HxnjNGnKn2Y?list=PLo_tFZGoEZum8b35jClphx5kL8AP1bGut

Xiao nRF52840 Sense IMU使用方法

seeed studio

快速入口 ▾ 按主题探索 ▾ 常见问题 ▾ 参与我们 ▾

CN 简体中文 ▾

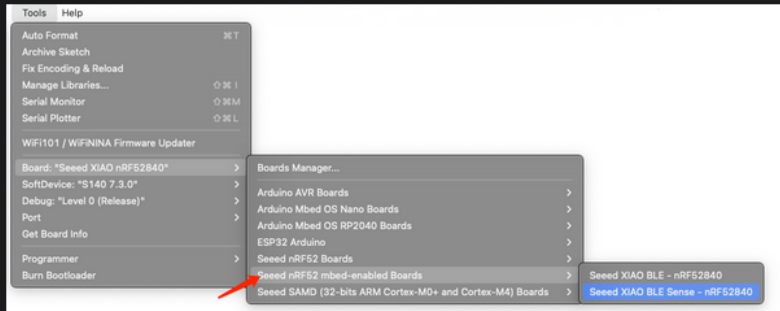
- XIAO MG24 >
- XIAO RP2040 >
- XIAO RP2350 >
- XIAO nRF52840 系列 ▾
 - Seeed Studio XIAO nRF52840 系列入门指南
 - RTOS >
 - 编程语言 >
 - 平台 >
 - 硬件使用 ▾
 - XIAO nRF52840 Sense 的 IMU 使用方法**
 - XIAO nRF52840 Sense 的 PDM 使用方法
 - XIAO nRF52840 Sense 的 QSPI Flash
 - 两个版本的引脚复用
 - 两个版本的 NFC 使用
 - 蓝牙库 >
 - 嵌入式机器学习 >
 - 应用 >

Seeed Studio XIAO nRF52840 Sense 上的 6 轴 IMU 使用方法

Seeed Studio XIAO nRF52840 Sense 配备了一个高精度的 6 轴惯性测量单元 (IMU)，包括一个 3 轴加速度计 和一个 3 轴陀螺仪。该模块上还有一个 内置温度传感器。我们相信这个模块可以在您的 TinyML 项目中提供很大帮助。本教程将介绍在该开发板上使用此 IMU 的基础知识。

注意

- Seeed Studio XIAO nRF52840 没有配备此 IMU 模块。
- 当我们使用 "Seeed nrf52 mbed-enabled Boards Library" 时，IMU 功能会表现得更好，因此我们强烈推荐使用它。

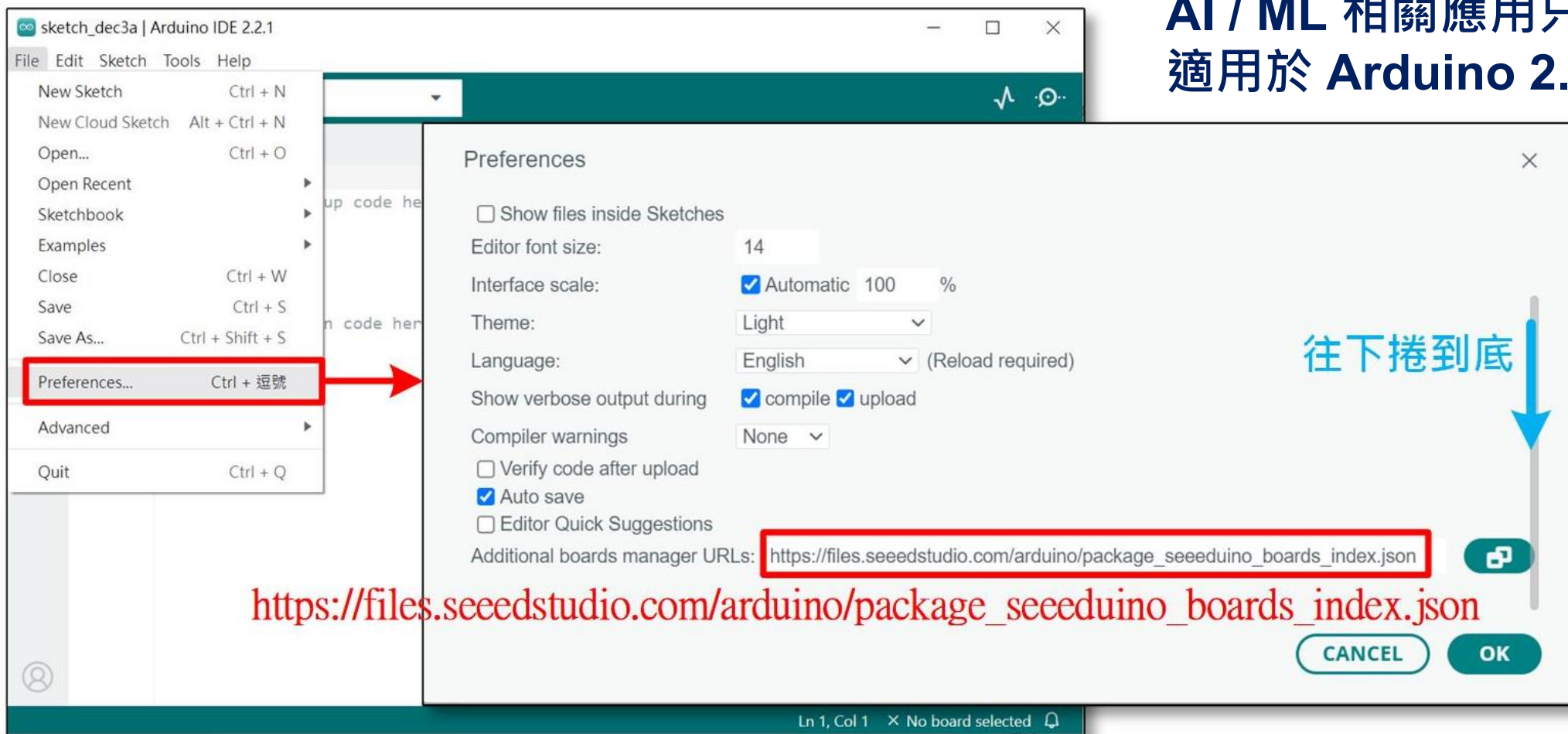


在串口监视器上查看加速度计、陀螺仪和温度数据

<https://wiki.seeedstudio.com/cn/XIAO-BLE-Sense-IMU-Usage/>

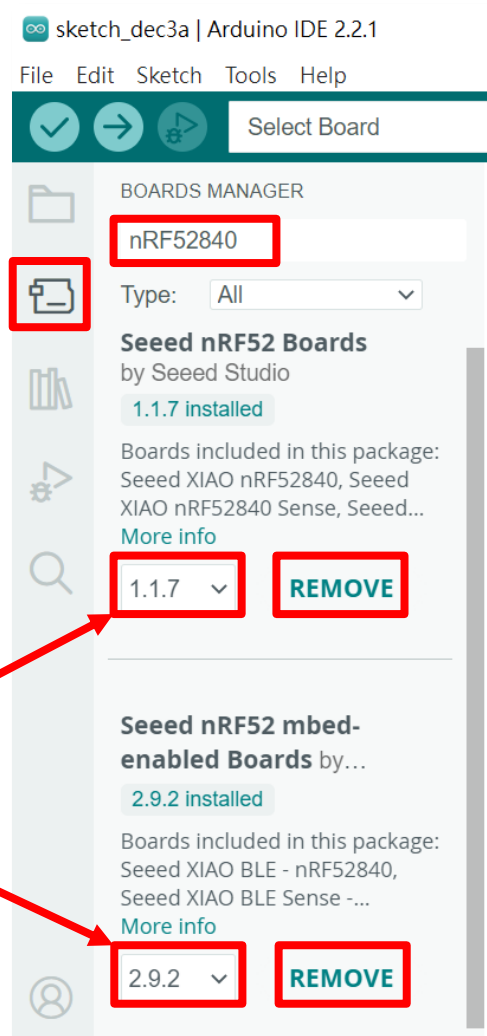
Arduino 新增開發板設定

AI / ML 相關應用只
適用於 Arduino 2.x 。



OmniXRI整理製作, 2023/12/08

安裝Seeed nRF52840函式庫



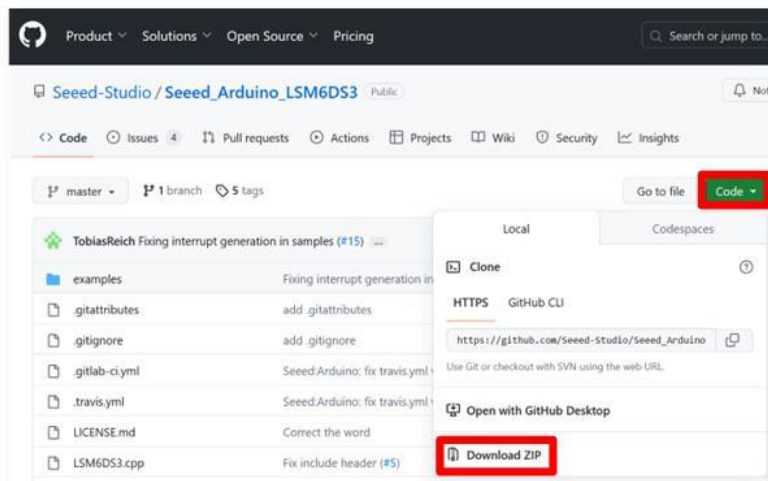
- 點選選單 Tools > Board > Boards Manager...，或直接點選左側第二個「開發板」圖示。
- 輸入 **nRF52840** 搜尋 Seeed nRF52840 開發板相關函式庫。
- 點選「**INSTALL**」安裝下列二個函式庫。（版本可取最新的）
 - **Seeed nRF52 Boards**（BLE, 低功耗功能）
 - **Seeed nRF52 mbed-enabled Boards**（PDM, IMU, ML）
- 安裝後若不需要時，可點選「**REMOVE**」解除安裝。



16

安裝運動感測器 (IMU) 函式庫

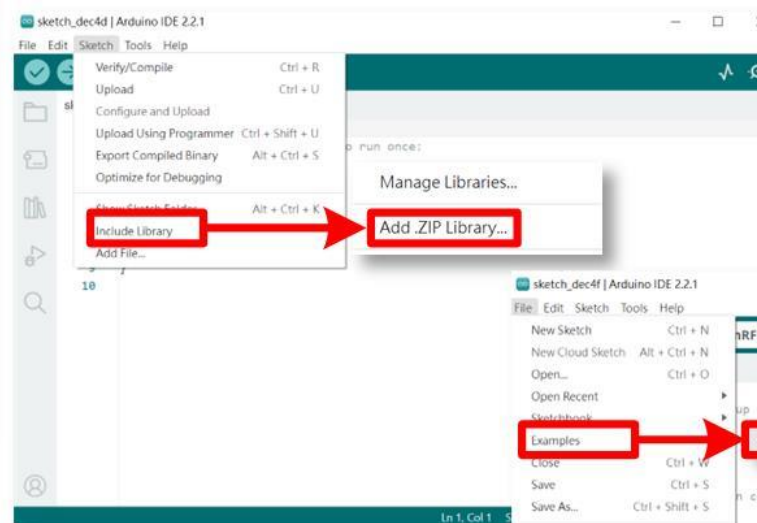
https://github.com/Seeed-Studio/Seeed_Arduino_LSM6DS3



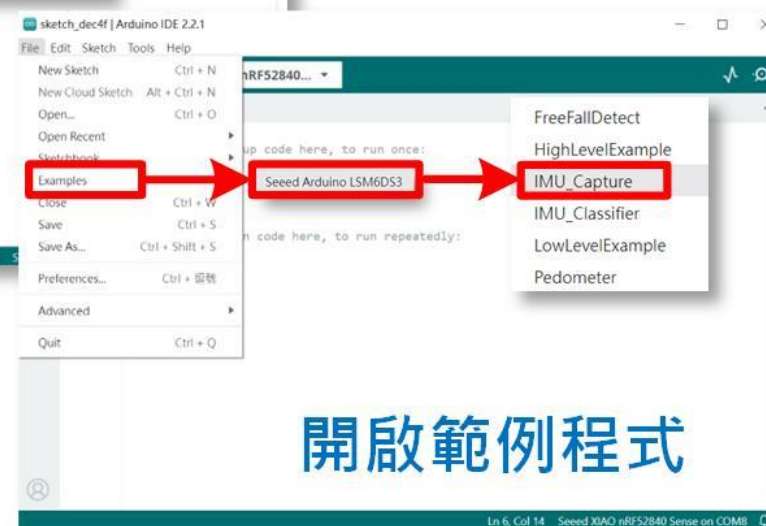
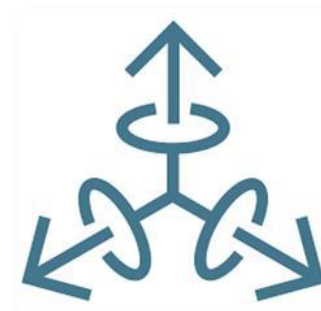
Seeed_Arduino_LSM6DS3-master.zip
下載運動感測器函式庫



https://github.com/Seeed-Studio/Seeed_Arduino_LSM6DS3



安裝函式庫



開啟範例程式

OmniXRI整理製作, 2023/12/08

運動感測器 (IMU) 取值範例

File > Examples > Seeed Arduino LSM6DS3 > IMU_capture

```
#include <LSM6DS3.h>
#include <Wire.h>
LSM6DS3 myIMU(I2C_MODE, 0x6A); // 建立LSM6DS3實例並設定I2C裝置位址
float aX, aY, aZ, gX, gY, gZ; // 儲存三軸加速度及陀螺儀數值變數
const float accelerationThreshold = 2.5; // 加速度門檻值
const int numSamples = 119; // 樣本數量
int samplesRead = numSamples; // 讀取樣本數量
void setup() { // 初始化設定
    Serial.begin(9600); // 設定串列通信速度(bps)
    while (!Serial); // 等待串列埠初始化完成
    if (myIMU.begin() != 0) { // 初始化運動感測器
        Serial.println("Device error");
    } else {
        Serial.println("aX,aY,aZ,gX,gY,gZ");
    }
}
void loop() { // 無限迴圈
    // 等待取得足夠數量樣本
    while (samplesRead == numSamples) {
        // 讀取加速度計值
        aX = myIMU.readFloatAccelX();
        aY = myIMU.readFloatAccelY();
        aZ = myIMU.readFloatAccelZ();
        // 求得加速度值總合
        float aSum = fabs(aX) + fabs(aY) + fabs(aZ);
        // 檢查是否超過門檻值，若是重置已讀樣本數量
        if (aSum >= accelerationThreshold) {
            samplesRead = 0;
            break;
        }
    }
}
```

觸發值大於
門檻才輸出

```
// 檢查自上次檢測到顯著運動以來是否已讀取所有必需的樣本
while (samplesRead < numSamples) {
    samplesRead++; // 已讀取樣本數加1
    // print the data in CSV format
    Serial.print(myIMU.readFloatAccelX(), 3);
    Serial.print(',');
    Serial.print(myIMU.readFloatAccelY(), 3);
    Serial.print(',');
    Serial.print(myIMU.readFloatAccelZ(), 3);
    Serial.print(',');
    Serial.print(myIMU.readFloatGyroX(), 3);
    Serial.print(',');
    Serial.print(myIMU.readFloatGyroY(), 3);
    Serial.print(',');
    Serial.print(myIMU.readFloatGyroZ(), 3);
    Serial.println();
    // 若已讀樣本數已足夠則列印換行
    if (samplesRead == numSamples) {
        Serial.println();
    }
}
```

練習：將板子以不同方向移動及旋轉測試讀值是否正常

IMU 連續取值輸出至 Edge Impulse

```

1  #include <LSM6DS3.h>
2  #include <Wire.h>
3
4  //Create a instance of class LSM6DS3
5  LSM6DS3 myIMU(I2C_MODE, 0x6A);    //I2C device address 0x6A
6  float aX, aY, aZ, gX, gY, gZ;
7  // const float accelerationThreshold = 2.5; // threshold of significant in G's
8  // const int numSamples = 119;
9  // int samplesRead = numSamples;
10
11 void setup() {
12     // put your setup code here, to run once:
13     Serial.begin(115200);
14     while (!Serial);
15     //Call .begin() to configure the IMUs
16     if (myIMU.begin() != 0) {
17         Serial.println("Device error");
18     } else {
19         Serial.println("aX,aY,aZ,gX,gY,gZ");
20     }
21 }
22
23 void loop() {
24     // wait for significant motion
25     // while (samplesRead == numSamples) {
26     //     // read the acceleration data
27     //     aX = myIMU.readFloatAccelX();
28     //     aY = myIMU.readFloatAccelY();
29     //     aZ = myIMU.readFloatAccelZ();
30
31     //     // sum up the absolutes
32     //     float aSum = fabs(aX) + fabs(aY) + fabs(aZ);
33
34     //     // check if it's above the threshold

```

註解或刪除
紅色框區域

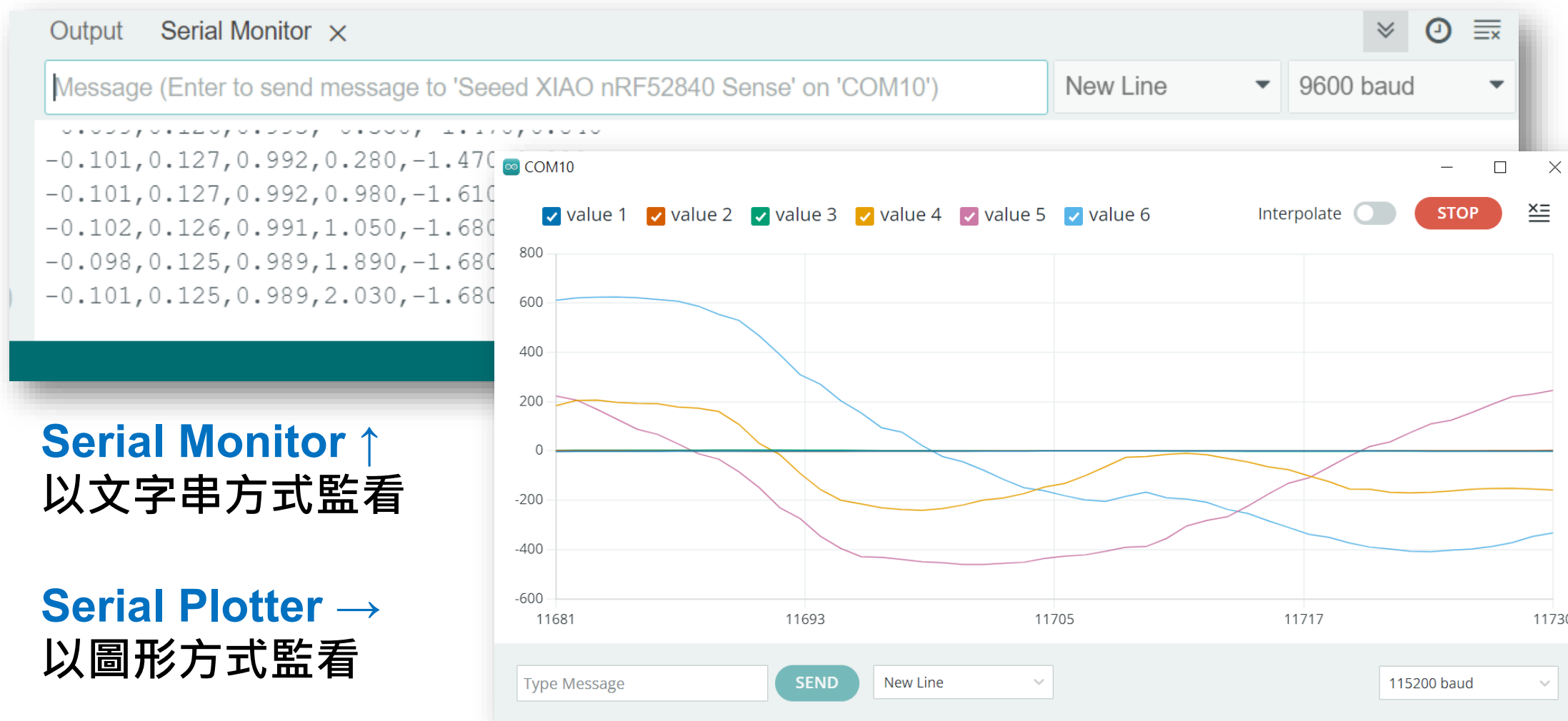
```

35 // if (aSum >= accelerationThreshold) {
36 //     // reset the sample read count
37 //     samplesRead = 0;
38 //     break;
39 // }
40 // }
41
42 // // check if the all the required samples have been read since
43 // // the last time the significant motion was detected
44 // while (samplesRead < numSamples) {
45 //     // check if both new acceleration and gyroscope data is
46 //     // available
47 //     // read the acceleration and gyroscope data
48
49 //     samplesRead++;
50
51 // print the data in CSV format
52 Serial.print(myIMU.readFloatAccelX(), 3);
53 Serial.print(',');
54 Serial.print(myIMU.readFloatAccelY(), 3);
55 Serial.print(',');
56 Serial.print(myIMU.readFloatAccelZ(), 3);
57 Serial.print(',');
58 Serial.print(myIMU.readFloatGyroX(), 3);
59 Serial.print(',');
60 Serial.print(myIMU.readFloatGyroY(), 3);
61 Serial.print(',');
62 Serial.print(myIMU.readFloatGyroZ(), 3);
63 Serial.println();
64
65 // if (samplesRead == numSamples) {
66 //     // add an empty line if it's the last sample
67 //     Serial.println();
68 // }
69 // }
70 }

```

連續取值並輸出至 COM
搭配 edge-impulse-
data-forwarder 可將感
測器值送到雲端

監看運動感測器 (IMU) 輸出值



Serial Monitor ↑
以文字串方式監看

Serial Plotter →
以圖形方式監看

案例分享 — Rabboni



bboni Ai
Edge AI
Development Kit
Withwe2 ANDRabboni

陽明交大電機工程系溫瓊岸教授團隊衍生新創公司產品，內建六軸重力感測器、藍芽傳輸及運算元件，可即時傳輸感測讀值並提供取樣頻率及動態範圍之多樣選擇。

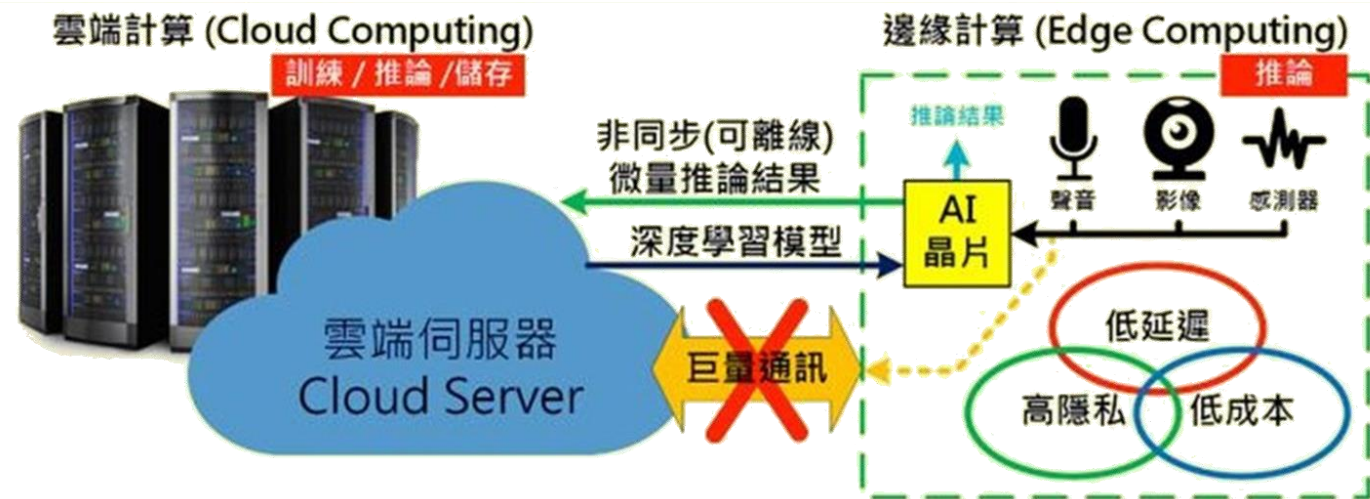
資料來源：<https://rabboni.com.tw/>



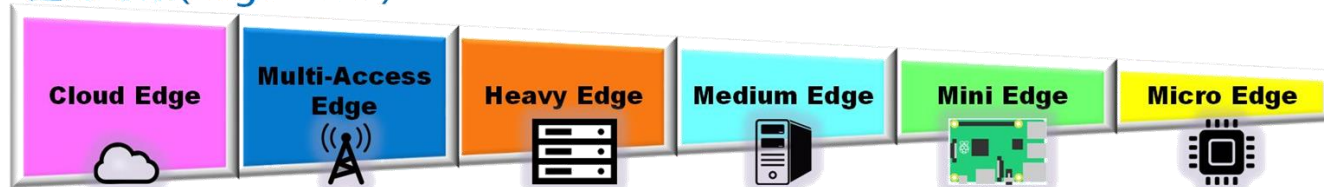
2025 陽明交大攜手奇景光電推動AI 應用創新
菁英培育計畫決賽暨成果發表會



Edge AI



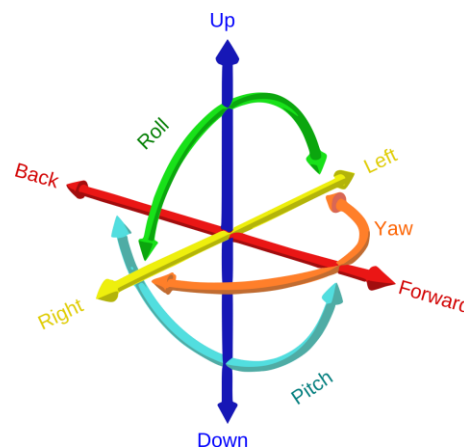
邊緣等級(Edge Level)



9.2. 運動手勢辨識

設計自定義操作手勢

- Start — 向前張開手掌
 - Next — 由左向右快速平移並有頓點
 - Prev — 由下向上擺動並有頓點
 - Stop — 雙手快速交叉於胸口
 - Unknow — 任意輕微擺動或不動，可為任意姿勢。
- 運動感測器必須固定在手部特定位置以免相同動作但感測值差異過大
 - 動作設定最好於不同軸向有明顯差異可只用加速度計而不使用陀螺儀

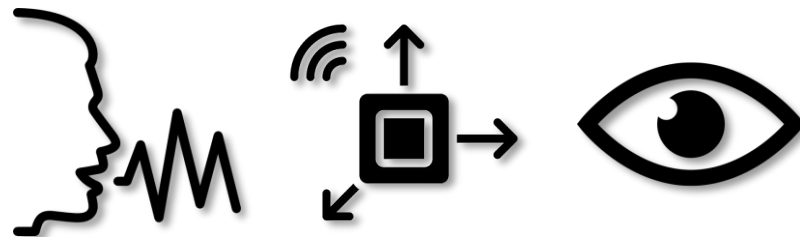


TinyML 開發流程選項

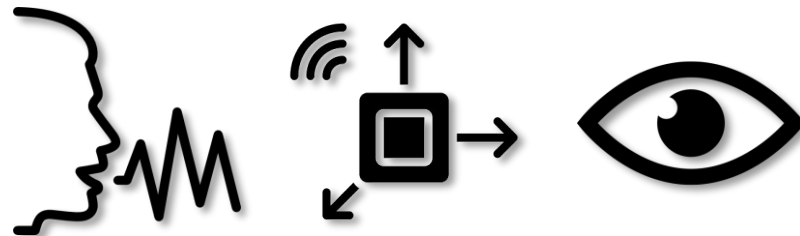
模型訓練工具名稱

適用感測器

主要特色



免費版：資料集需自行建置，僅提供模型訓練及轉成 TensorFlow Lite for Microcontroller 格式。可利用雲端 Google Colab 或本地端 Jupyter Notebooks 進行模型訓練及部署。

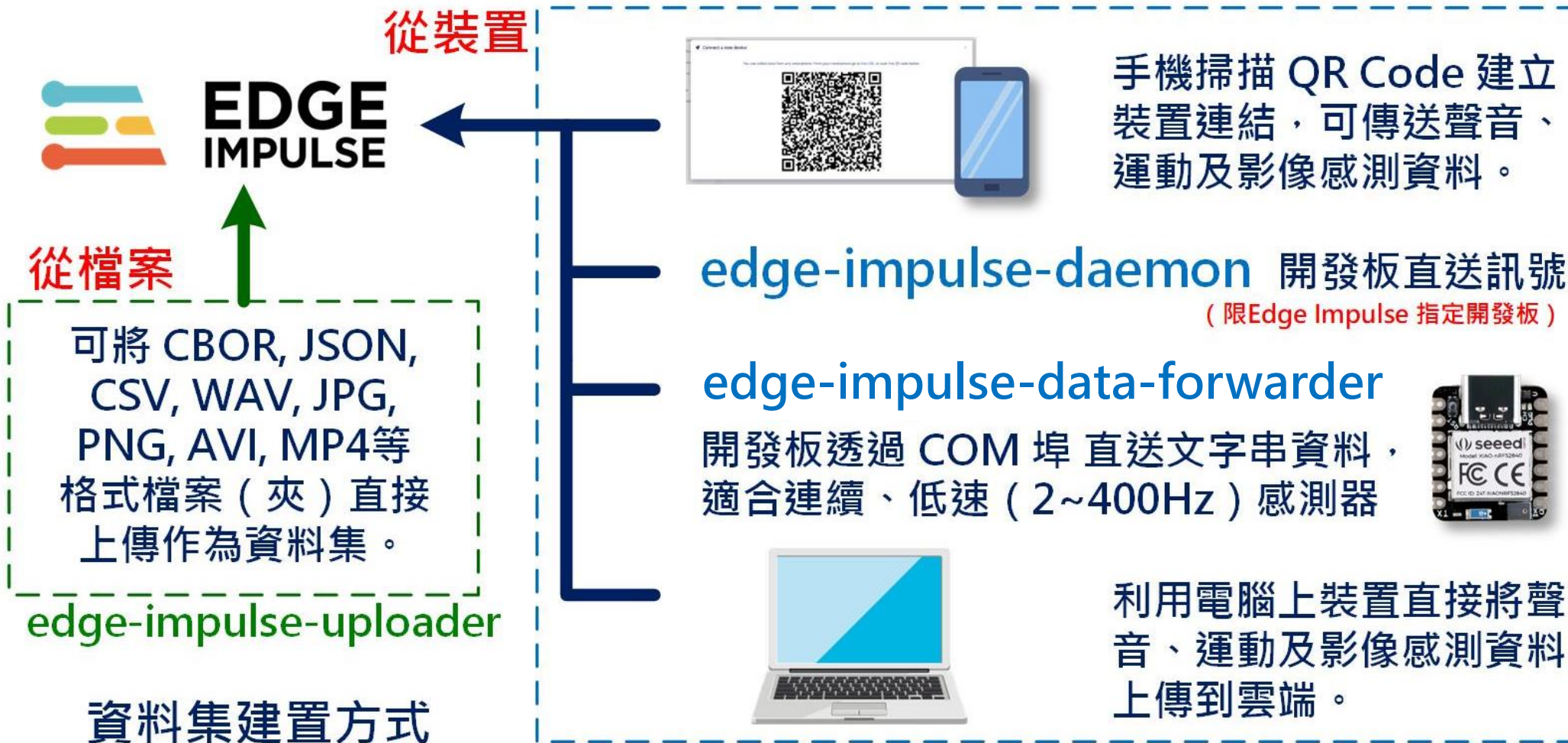


免費版：可支援各式感測器及開發板，提供一定額度資料集建置、資料前處理（特徵提取）模型訓練、模型優化及多種部署方式。亦可導入已訓練好的 TensorFlow Lite, ONNX 等模型。



免費版：影像分類可多分類、物件偵測同一模型僅能一類。事件觸發只能設一種條件。可支援雲端資料收集、模型訓練及部署，亦導入外部已訓練好之 TensorFlow Lite 模型。

Edge Impulse 資料集建置方式



OmniXRI整理製作, 2025/04/16

快速操作指令表



https://github.com/OmniXRI/Edge_AI_TinyML_Course_2024/blob/main/Ch13_Motion_Recognition/IMU_Quick_Guide.md

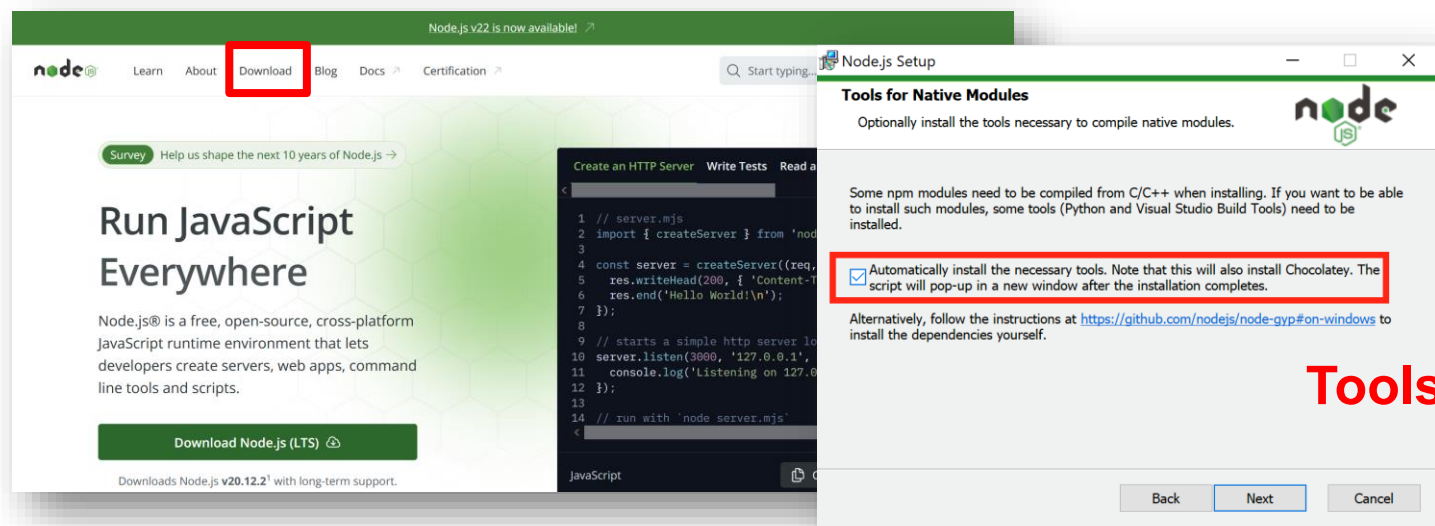
下載及安裝必要工具

1. Python 3.x



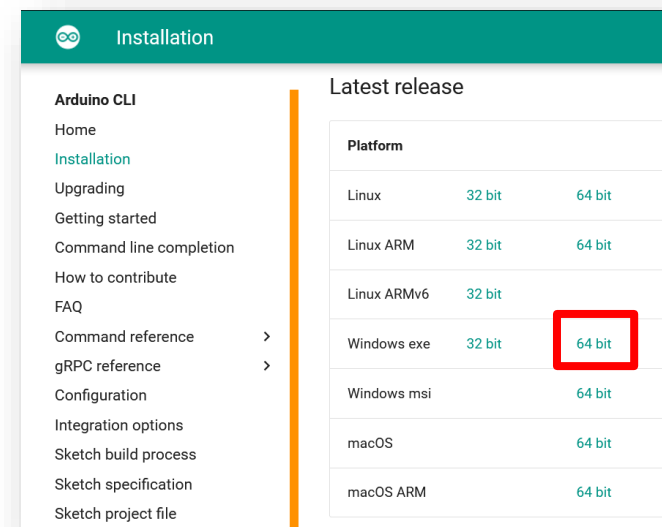
2. Node.js

(若後續 Edge Impulse 無法安裝
可換回舊版，例：v18.20.2 LTS)



3. Arduino CLI

(選配，依不同開發板需求)



安裝時要注意
記得一定要勾選

Tools for Native Modules

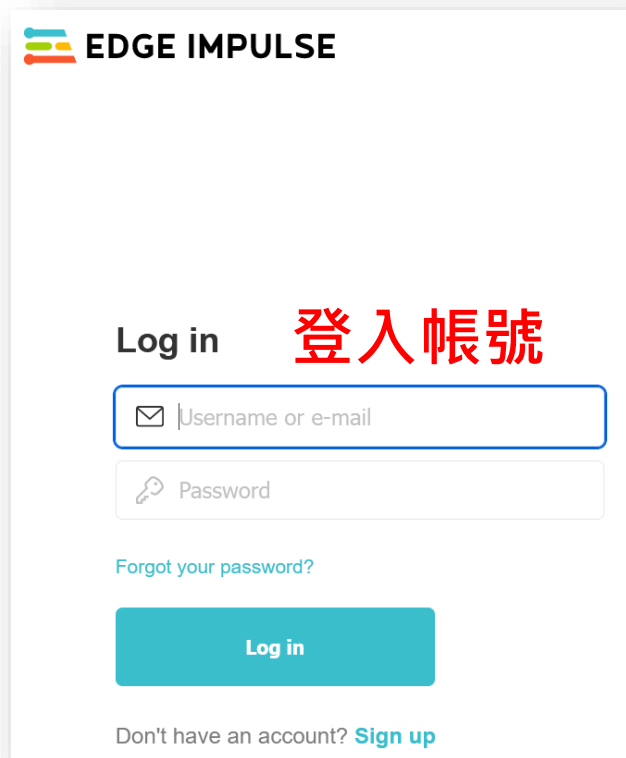
以 `node -v` 及 `npm -v`
檢查安裝好的版本

安裝 Edge Impulse windows 工作環境

- 安裝**Edge Impulse CLI**
`npm install -g edge-impulse-cli --force`
- 安裝完成後會得到下列工具
 - **edge-impulse-daemon** 透過 **COM** 埠和雲端平台連接
 - **edge-impulse-uploader** 允許上傳本機檔案
 - **edge-impulse-data-forwarder** 透過 **COM** 埠上傳開發板感測器數值
 - **edge-impulse-run-impulse** 顯示裝置上運行的模型
 - **edge-impulse-blocks** 建立自定義區塊
 - **himax-flash-tool** 燒錄 Haimax WE-I Plus

<https://docs.edgeimpulse.com/docs/tools/edge-impulse-cli/cli-installation>

建立 Edge Impulse 新專案



EDGE IMPULSE

Log in **登入帳號**

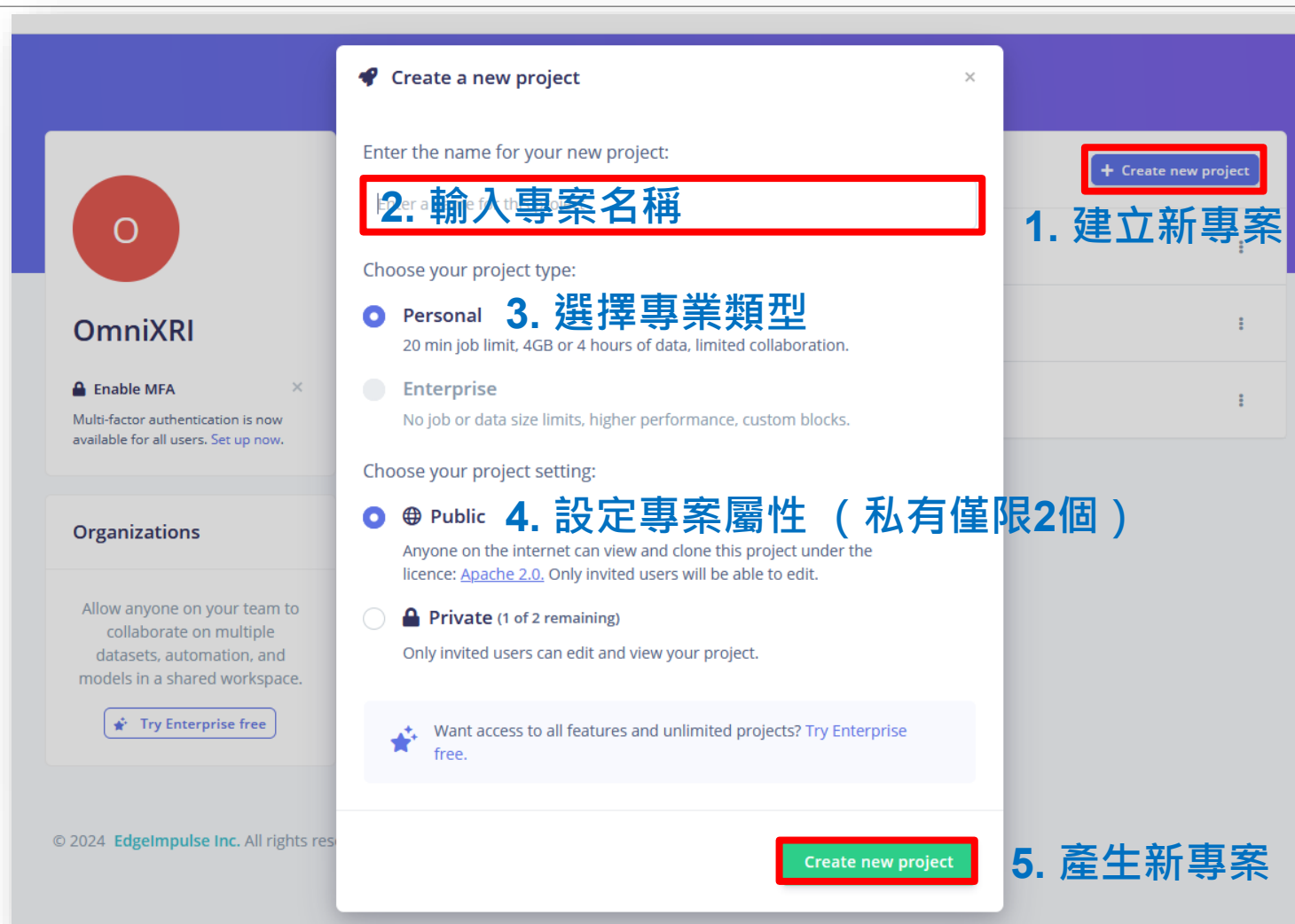
Username or e-mail

Password

Forgot your password?

Log in

Don't have an account? [Sign up](#)



Create a new project

Enter the name for your new project:

2. 輸入專案名稱

Choose your project type:

☒ **Personal** **3. 選擇專業類型**
20 min job limit, 4GB or 4 hours of data, limited collaboration.

☐ Enterprise
No job or data size limits, higher performance, custom blocks.

Choose your project setting:

☒ **Public** **4. 設定專案屬性 (私有僅限2個)**
Anyone on the internet can view and clone this project under the licence: [Apache 2.0](#). Only invited users will be able to edit.

☐ Private (1 of 2 remaining)
Only invited users can edit and view your project.

Want access to all features and unlimited projects? [Try Enterprise free.](#)

Create new project

1. 建立新專案

5. 產生新專案

參考資料：<https://wiki.seeedstudio.com/cn/XIAO-BLE-PDM-EI/>

啟動edge-impulse-data-forwarder

```

C:\WINDOWS\system32\cmd.exe - "node" "C:\Users\jack\AppData\Roaming\npm\node_modules\edge-impulse-cli\build\cli\data-forward...
C:\Users\jack>edge-impulse-data-forwarder --clean edge-impulse-data-forwarder --clean
Edge Impulse data forwarder v1.24.1
? What is your user name or e-mail address (edgeimpulse.com)? omnixri@gmail.com 輸入帳號
? What is your password? [hidden] 輸入密碼
Endpoints:
  Websocket: wss://remote-mgmt.edgeimpulse.com
  API:       https://studio.edgeimpulse.com
  Ingestion: https://ingestion.edgeimpulse.com

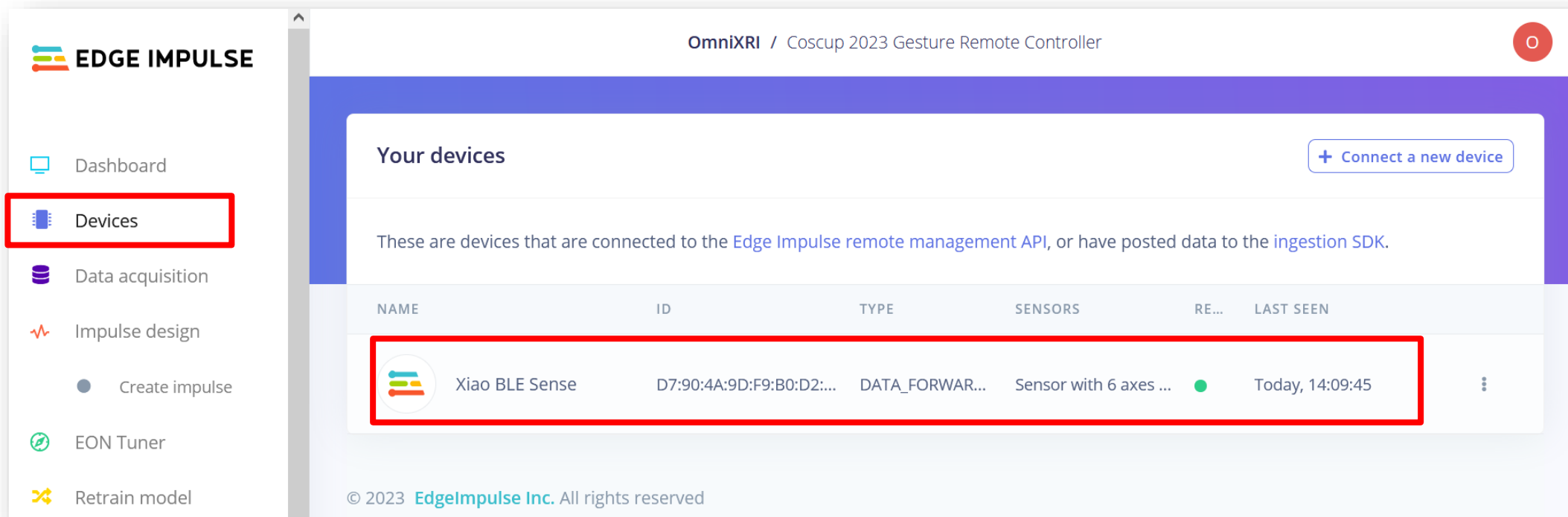
[SER] Connecting to COM8
[SER] Serial is connected (7F:63:29:74:0B:F4:EA:69)
[WS ] Connecting to wss://remote-mgmt.edgeimpulse.com
[WS ] Connected to wss://remote-mgmt.edgeimpulse.com

? To which project do you want to connect this device? OmniXRI / Xiao_nRF52840_IMU_Test 選擇對應專案
[SER] Detecting data frequency...
[SER] Detected data frequency: 130Hz
? 6 sensor axes detected (example values: [-0.02,-0.005,1.001,0.49,-1.4,0.84]). What do you want to call them? Separate
the names with ',': aX,aY,aX,gX,gY,gZ 輸入資料數量及名稱
[WS ] Device "XiaoSense" is now connected to project "Xiao_nRF52840_IMU_Test". To connect to another project, run `edge-
impulse-data-forwarder --clean`.
[WS ] Go to https://studio.edgeimpulse.com/studio/391754/acquisition/training to build your machine learning model!
[WS ] Incoming sampling request {
  path: '/api/training/data',
  label: 'OK',
  length: 10000,
  interval: 7.6923076923076925,
  hmacKey: 'd00f5918c1020b658b18569f9ee881c5',
  sensor: 'Sensor with 6 axes (aX, aY, aX, gX, gY, gZ)'
}

```

輸入帳號密碼並選定專案。準備接收來自開發板傳送的資料並給予並應的名稱。傳送的資料頻度必須大於**2Hz**。
若無法連接 **COM** 埠，請按開發板重置鍵一次，並請關閉 **Arduino** 及有可能佔用 **COM** 之程式。

檢查裝置是否已連線



- 開發板運行IMU連續取值輸出程式。
- 啟動 **edge-impulse-data-forwarder**，將裝置連線到雲端。
- 將開發板輸出之內容，傳送到雲端。

從外部裝置取得資料

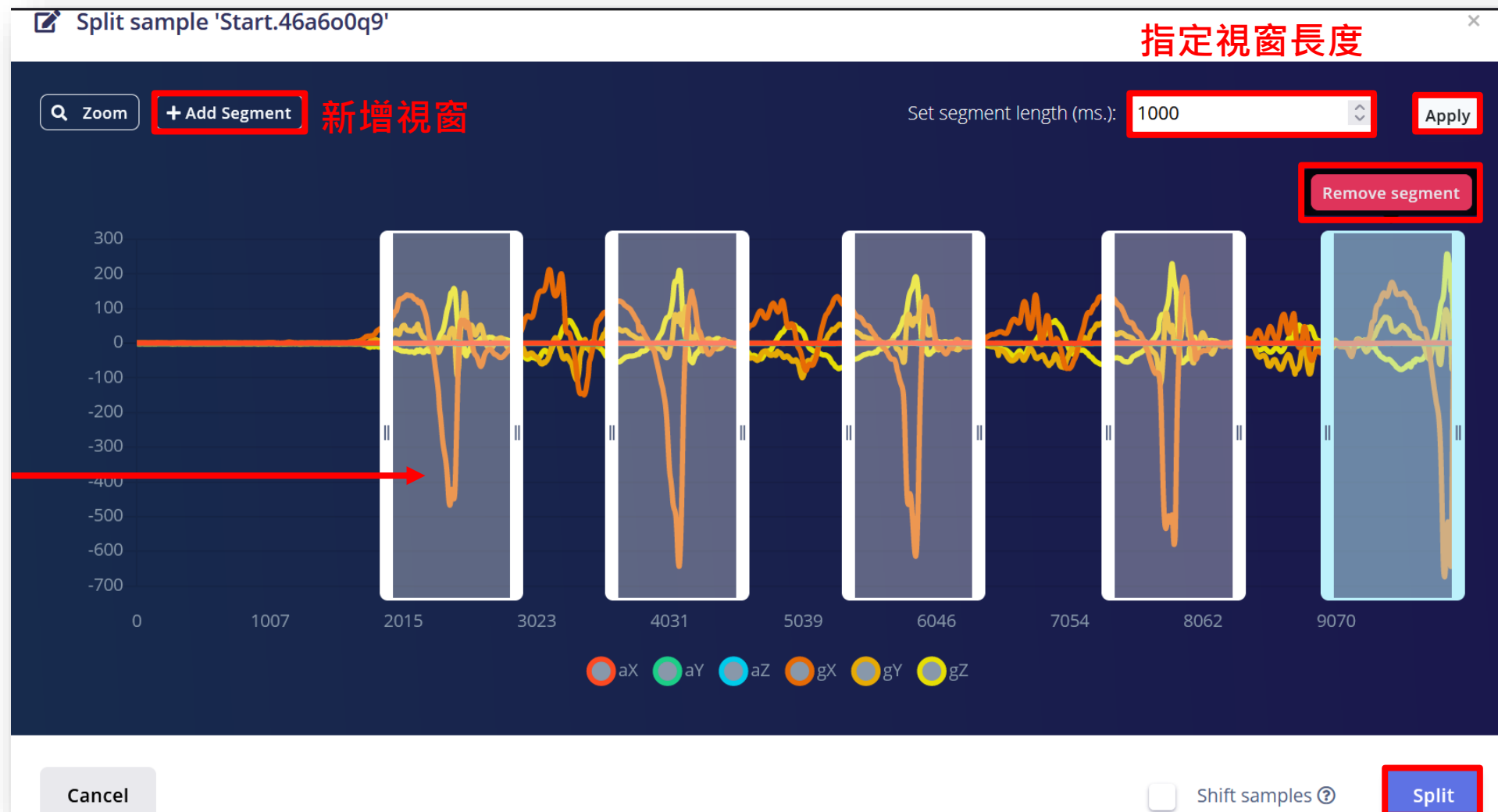
The screenshot displays the Edge Impulse OmniXRI web interface. On the left, a sidebar lists navigation options: Dashboard, Devices, **Data acquisition** (highlighted with a red box), Impulse design, Create impulse, EON Tuner, Retrain model, Live classification, Model testing, Versioning, and Deployment. The main content area is titled 'Dataset' and includes tabs for 'Dataset', 'Data explorer', 'Data sources', and 'CSV Wizard'. The 'Dataset' tab is active, showing an 'Add data' section with the text 'Start building your dataset by adding some data.' and a '+ Add data' button. To the right, the 'Collect data' section is visible, featuring a 'Device' dropdown menu set to 'Xiao BLE Sense'. Below this, there are fields for 'Label' (containing the text '指定標籤名稱' - Specify label name, highlighted with a red box), 'Sample length (ms.)' (containing '0.5s' and the text '取樣時長' - Sampling time, highlighted with a red box), 'Sensor' (set to 'Sensor with 6 axes (aX, aY, aZ, gX, gY)'), and 'Frequency' (set to '128Hz'). At the bottom right of the 'Collect data' section, there is a 'Start sampling' button (highlighted with a red box).

大量收集樣本並分割成獨立可訓練樣本

The screenshot displays the Edge Impulse web interface. On the left, a sidebar contains navigation links: Dashboard, Devices, **Data acquisition** (highlighted with a red box), Impulse design, EON Tuner, Retrain model, Live classification, Model testing, Versioning, and Deployment. The main panel is divided into two sections. The top section, titled 'Dataset', shows 'DATA COLLECTED 10s' and a table with columns 'SAMPLE NAME' and 'LABEL'. The table contains one row: 'Start.46a6o0q9' with label 'Start'. A context menu is open over this row, listing actions: Rename, Edit label, Move to test set, Disable, Crop sample, **Split sample** (highlighted with a red box), Download, and Delete. The text '資料分割' (Data Splitting) is written in red next to the 'Split sample' option. The bottom section of the main panel shows 'Training (1)' and 'Test (0)' counts. The right panel, titled 'Collect data', includes settings for 'Device' (Xiao BLE Sense), 'Label' (Start), 'Sample length (ms.)' (10000), 'Sensor' (Sensor with 6 axes), and 'Frequency' (128Hz). A 'Start sampling' button is at the bottom right.

原始資料與自動分割

按住滑鼠
左鍵不放
可左右拖
拉移動



自動分割
刪除視窗

完成分割

反復收集分割，建立完整資料集

The screenshot displays the Edge Impulse web interface. On the left, a sidebar contains navigation options: Dashboard, Devices, Data acquisition (highlighted with a red box), Impulse design, EON Tuner, Retrain model, Live classification, and Model testing. The main area is titled 'OmniXRI / Coscup 2023 Gesture Remote Controller' and features tabs for Dataset, Data explorer, Data sources, and CSV Wizard. A 'DATA COLLECT...' widget shows a progress bar and a pie chart, with a red box highlighting it. Below this, a 'Dataset' table lists training and test samples. A red text annotation '標籤數量及比例' (Label quantity and ratio) points to the 'Training' and 'Test' columns. The table has columns for Sample Name, Label, Added, and Length. The 'Collect data' section on the right includes settings for Device (Xiao BLE Sense), Label (unknown), Sample length (10000 ms), Sensor (Sensor with 6 axes), and Frequency (128Hz), with a 'Start sampling' button.

Dataset

DATA COLLECT... 3m 11s

TRAIN / TEST 100...

Dataset 標籤數量及比例

Training (191)		Test (0)	
SAMPLE NAME	LABEL	ADDED	LENGTH
unknown.46a8r...	unknown	Today, 14:5...	1s
unknown.46a8r...	unknown	Today, 14:5...	1s
unknown.46a8r...	unknown	Today, 14:5...	1s
unknown.46a8r...	unknown	Today, 14:5...	1s

Collect data

Device ?
Xiao BLE Sense

Label
unknown

Sample length (ms.)
10000

Sensor
Sensor with 6 axes (aX, aY, aZ, gX, gY)

Frequency
128Hz

Start sampling

選擇模型及設定必要參數

EDGE IMPULSE

- Dashboard
- Devices
- Data acquisition
- Impulse design
 - Create impulse**
 - Spectral features
 - Classifier
- EON Tuner
- Retrain model
- Live classification
- Model testing

Time series data

Input axes (6)
aX, aY, aZ, gX, gY, gZ

Window size
1000 ms.

Window increase
200 ms.

Frequency (Hz)
128

Zero-pad data
☒

Spectral Analysis

Name
Spectral features

Input axes (3)
☒ aX
☒ aY
☒ aZ
☐ gX
☐ gY
☐ gZ

Classification

Name
Classifier

Input features
☒ Spectral features

Output features
4 (Next, Prev, Start, Stop)

Output features

4 (Next, Prev, Start, Stop)

Save Impulse

可選擇性只保留加速度計值

可依需求自行新增修改 Block

修改後記得存檔

提取資料特徵

The image displays two screenshots of the Edge Impulse web interface, illustrating the steps to generate features from raw data.

Left Screenshot: The interface shows the 'OmniXRI / Coscup 2023 Gesture Remote Controller' project. The 'Generate features' button is highlighted with a red box. The 'Spectral features' option in the left sidebar is also highlighted with a red box. The 'Raw data' section shows a waveform plot. Below it, the 'Raw features' section displays a list of feature values: `-0.4640, -0.2830, -0.8090, -0.4350,`.

Right Screenshot: The interface shows the 'Training set' configuration. The 'Generate features' button is highlighted with a red box. The 'Feature explorer' section shows 'No features generated yet.' The 'Training set' configuration includes:

- Data in training set: 2m 0s
- Classes: 4 (Next, Prev, Start, Stop)
- Training windows: 120
- Calculate feature importance: ☐

The 'Generate features' button is highlighted with a red box, and the text '產生特徵值' (Generate feature values) is written next to it.

提取特徵結果

Feature generation output

🔊 (0)

Still running...

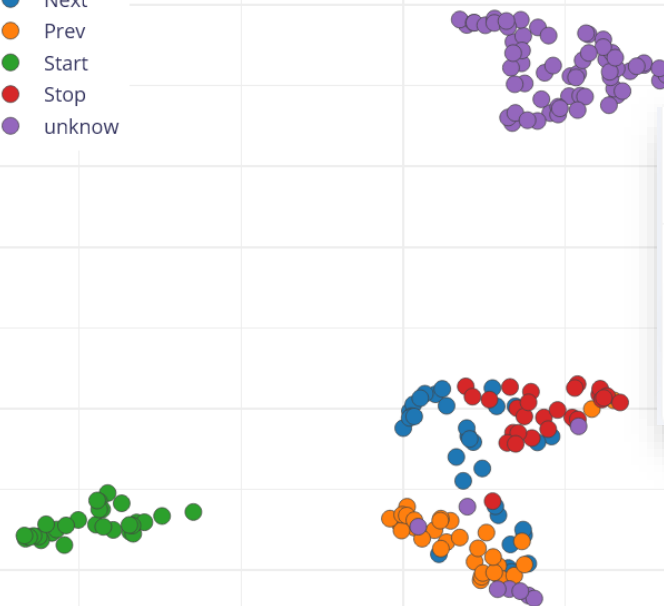
```
completed 0 / 500 epochs
completed 50 / 500 epochs
completed 100 / 500 epochs
completed 150 / 500 epochs
completed 200 / 500 epochs
completed 250 / 500 epochs
completed 300 / 500 epochs
completed 350 / 500 epochs
completed 400 / 500 epochs
completed 450 / 500 epochs
```

Fri Jul 28 06:54:26 2023 Finished embedding
Reducing dimensions for visualizations OK

Job completed

Feature explorer

- Next
- Prev
- Start
- Stop
- unknown



On-device performance ⓘ



PROCESSING TIME
23 ms.



PEAK RAM USAGE
2 KB

以指定的硬體進行資源
使用及推論時間估測

設定分類訓練相關參數

The screenshot displays the Edge Impulse web interface for configuring a classification model. The left sidebar shows the navigation menu with 'Classifier' selected. The main area is divided into several sections:

- Neural Network settings:** Includes a description field and a target device dropdown set to 'Arduino Nano 33 BLE Sense (Cortex-M4F 64MHz)'.
- Training settings:**
 - Number of training cycles: 30
 - Learning rate: 0.0005
- Advanced training settings:**
 - Validation set size: 20 %
 - Split train/validation set on metadata key: (empty)
 - Auto-balance dataset: ☐
 - Profile int8 model: ☒
- Neural network architecture:**
 - Input layer (39 features)
 - Dense layer (20 neurons)
 - Dense layer (10 neurons)
 - Add an extra layer (button)
 - Output layer (5 classes)

At the bottom right, a green 'Start training' button is highlighted with a red box, accompanied by the text '開始訓練'.

訓練次數
學習率

**可勾選啟動GPU
加快訓練速度**

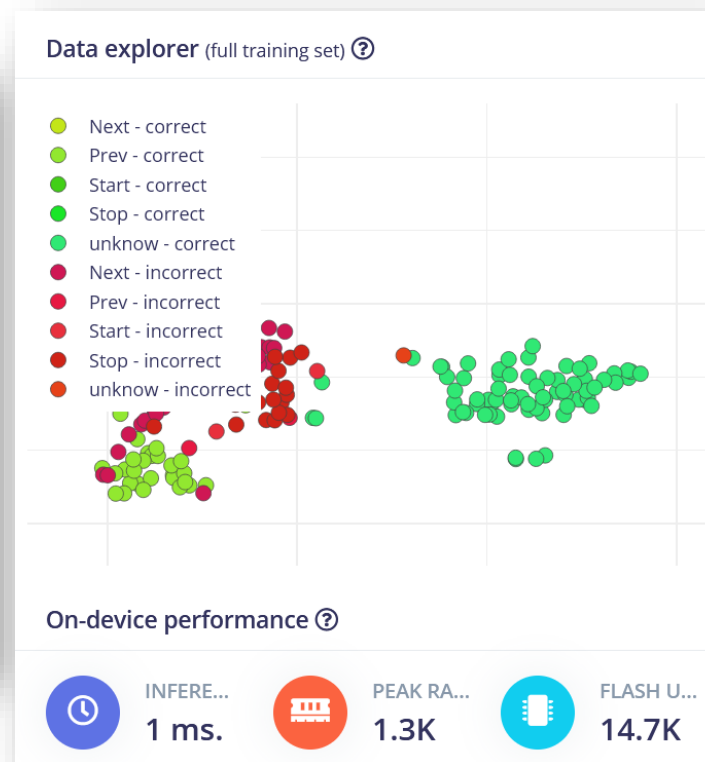
開始進行模型訓練及結果顯示

Training output ⚡ CPU (0) ▲

```
INFO: Created TensorFlow Lite XNNPACK delegate for CPU.
Calculating inferencing time OK
Profiling float32 model...
Profiling float32 model (TensorFlow Lite Micro)...
Profiling float32 model (EON)...
Attached to job 11100469...
Attached to job 11100469...
Profiling int8 model...
Profiling int8 model (TensorFlow Lite Micro)...
Profiling int8 model (EON)...
Attached to job 11100469...
Attached to job 11100469...

Model training complete

Job completed
```



Model Model version: ? Quantized (int8) ▼

Last training performance (validation set)

ACCURACY 82.1% LOSS 0.80

Confusion matrix (validation set)

	NEXT	PREV	START	STOP	UNKNOWN
NEXT	33.3%	50%	0%	16.7%	0%
PREV	0%	100%	0%	0%	0%
START	0%	0%	100%	0%	0%
STOP	20%	20%	0%	40%	20%
UNKNOWN	0%	0%	0%	0%	100%
F1 SCORE	0.44	0.75	1.00	0.50	0.97

Last training performance (validation set)

ACCURACY 94.9% LOSS 0.26

Confusion matrix (validation set)

	NEXT	PREV	START	STOP	UNKNOWN
NEXT	83.3%	16.7%	0%	0%	0%
PREV	0%	100%	0%	0%	0%
START	0%	0%	100%	0%	0%
STOP	20%	0%	0%	80%	0%
UNKNOWN	0%	0%	0%	0%	100%
F1 SCORE	0.83	0.92	1.00	0.89	1.00

**30
Epoch
結果**

**100
Epoch
結果**

線上測試（從外部裝置取樣）

The screenshot displays the Edge Impulse web interface for the 'OmniXRI / Xiao_nRF52840_KWS_Test' project. The left sidebar contains navigation links: Dashboard, Devices, Data acquisition, Impulse design, and a 'Try Enterprise Free' section. The main content area is divided into two panels. The 'Classify new data' panel on the left has four configuration options, each with a red annotation: 'Device' (指定連結裝置), 'Sensor' (指定感測器), 'Sample length (ms.)' (指定取樣時長), and 'Frequency' (指定取樣頻率). A green 'Start sampling' button is highlighted with a red box, with the text '開始取樣' next to it. The 'Classify existing test sample' panel on the right shows a dropdown menu set to 'white_noise (noise)' and a 'Load sample' button. The footer includes the copyright notice '© 2024 EdgeImpulse Inc. All rights reserved' and a help icon.

選擇部署種類及設定參數

EDGE IMPULSE

- Dashboard
- Devices
- Data acquisition
- Impulse design
 - Create impulse
 - Spectral features
 - Classifier
- EON Tuner
- Retrain model
- Live classification

Configure your deployment

You can deploy your impulse to any device. This makes the model run without an internet connection, minimizes latency, and runs with minimal power consumption. [Read more.](#)

Arduino library

Arduino library
An Arduino library with examples that runs on most Arm-based Arduino development boards.

MODEL OPTIMIZATIONS
Model optimizations can increase on-device performance but may reduce accuracy.

☒ Enable EON™ Compiler
Same accuracy, up to 50% less memory. [Learn more](#)

Quantized (int8)
Selected

	SPECTRAL	CLASSIFIER	TOTAL
LATENCY	23 ms.	1 ms.	24 ms.
RAM	2.2K	1.3K	2.2K
FLASH	-	14.7K	-
ACCURACY	-	-	-

Unoptimized (float32)
Select

	SPECTRAL	CLASSIFIER	TOTAL
LATENCY	23 ms.	27 ms.	50 ms.
RAM	2.2K	1.3K	2.2K
FLASH	-	13.6K	-
ACCURACY	-	-	-

To compare model accuracy, run model testing for all available optimizations. **Run model testing**

Estimate for Arduino Nano 33 BLE Sense (Cortex-M4F 64MHz) - [Change target](#)

Build

Latest build

ARDUINO v2 (Arduino library)
Today, 15:34:05 [View docs](#)

Build output

```

Container image pulled!
Job started
Writing templates...
Writing templates OK

Scheduling job in cluster...
Container image pulled!
Job started
Copying Edge Impulse SDK...
Copying Edge Impulse SDK OK

Compiling EON model...
Compiling EON model OK

Removing clutter and updating headers...
Removing clutter and updating headers OK

Creating archive...
Creating archive OK

Job completed
    
```

Built Arduino library

Add this library through the Arduino IDE via:

Sketch > Include Library > Add .ZIP Library...|

Examples can then be found under:

File > Examples > Xiao_nRF52840_IMU_Test_inferencing

以 **Arduino 函式庫(*.zip)**方式輸出
預設檔名為「**ei-專案名稱-arduino-版本序號.zip**」

導入 Arduino 函式庫並進行推論測試(1/2)

- 新增函式庫 **Sketch > Include Library > Add .ZIP Library ...**
- 新增範例 **File > Examples > ei-專案名稱_inferencing > nano_ble33_sense > nano_ble33_sense_accelerometer** (原始範例輸出結果從 **COM** 輸出文字串)
- 經編譯會在「**#error “Invalid model for current sensor”**」這列出現「**“invalid model for current sensor”**」，主要原因為Edge Impulse打包時是以Sensor_Fusion方式處理而不是Accelerometer方式造成。
- 點擊選單 **File > Preferences** 找到 Sketchbook Folder，進到該路徑後，進到 \libraries\專案名稱_inferencing\src\model-parameters\ 開啟 **model_metadata.h**。找到「**#define EI_CLASSIFIER_SENSOR EI_CLASSIFIER_SENSOR_FUSION**」，將定義改成「**EI_CLASSIFIER_SENSOR_ACCELEROMETER**」存檔即可。

導入 Arduino 函式庫並進行推論測試(2/2)

- 回到 Arduino 範例後會發現使用的運動感測器不同，所以需將「**#include <Arduino_LSM9DS1.h>**」註解掉並新增下列程式

```
#include <LSM6DS3.h>
```

```
#include <Wire.h>
```

```
LSM6DS3 myIMU(I2C_MODE, 0x6A);
```

- 在Setup函式中的初始化也要將
「if (!IMU.begin()) {」修改為
「if (!**myIMU**.begin()) {」

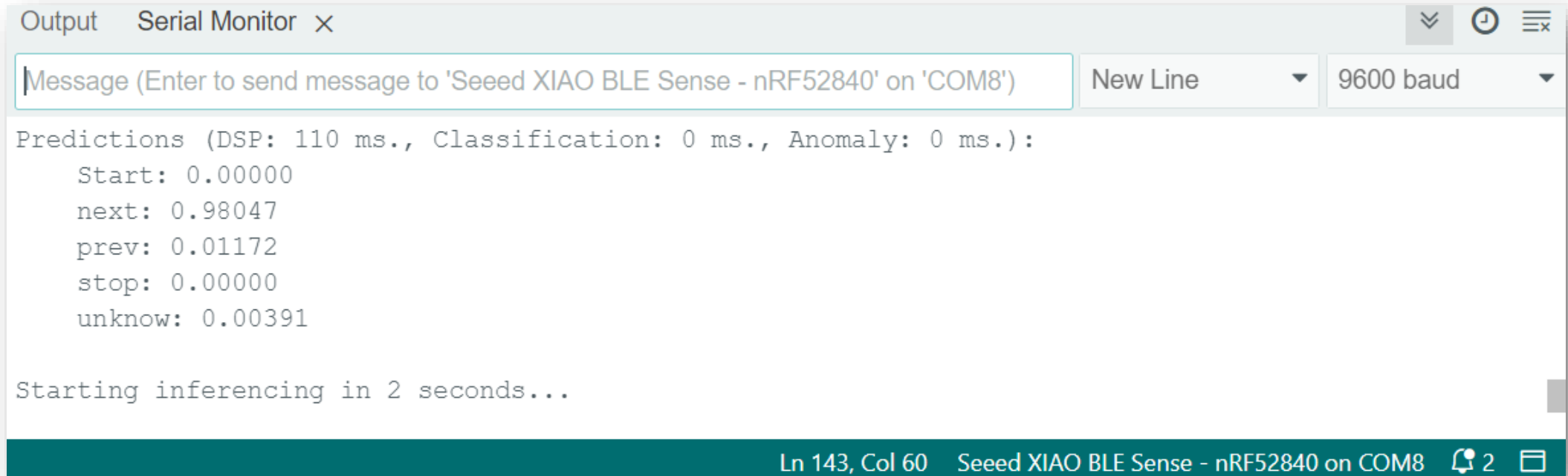
- 另外由於讀取IMU運動感測器函式不同因此也需修改如左。

- 重新編譯前確認開發板是設定在「**Seeed nRF52 mbed-enabled Boards** > Seeed XIAO BLE Sense - nRF52840」，不然程式運行後會找不到IMU LSM6DS3。

```
17  /* Includes -----
18  #include <Xiao_nRF52840_IMU_Test_inferencing.h>
19  // #include <Arduino_LSM9DS1.h> //Click here to get the libr
20  #include <LSM6DS3.h>
21  #include <Wire.h>
22  /* Constant defines -----
23  #define CONVERT_G_TO_MS2    9.80665f
24  #define MAX_ACCEPTED_RANGE  2.0f           // starting 03/2022
25
26  LSM6DS3 myIMU(I2C_MODE, 0x6A);    //I2C device address 0x6A
```

```
99  //      IMU.readAcceleration(buffer[ix], buffer[ix + 1],
100      buffer[ix]      = myIMU.readFloatAccelX();
101      buffer[ix + 1] = myIMU.readFloatAccelY();
102      buffer[ix + 2] = myIMU.readFloatAccelZ();
```

手勢辨識結果 (Arduino部份)



The screenshot shows the Arduino IDE Serial Monitor window. The title bar reads "Output Serial Monitor x". The input field contains the text "Message (Enter to send message to 'Seeed XIAO BLE Sense - nRF52840' on 'COM8')". The dropdown menu is set to "New Line" and the baud rate is "9600 baud". The output text displays the following information:

```
Predictions (DSP: 110 ms., Classification: 0 ms., Anomaly: 0 ms.):  
  Start: 0.00000  
  next: 0.98047  
  prev: 0.01172  
  stop: 0.00000  
  unknow: 0.00391  
  
Starting inferencing in 2 seconds...
```

The status bar at the bottom indicates "Ln 143, Col 60" and "Seeed XIAO BLE Sense - nRF52840 on COM8".

- 推論結果，包括前處理(DSP)、分類及異常時間 (ms)
- 將五個標籤的置信度顯示出來，1.0為完全正確，0.0為完全不正確，可設置門檻值來限制置信度不高之答案。
- **No Code型式**，範例程式直接編譯上傳，不需改寫任何程式，輸出結果由 **COM** 接收文字串。

Arduino 2.0 快速(增量)編譯



如何讓Arduino 2.0 快速編譯 (增量編譯)

非常重要：

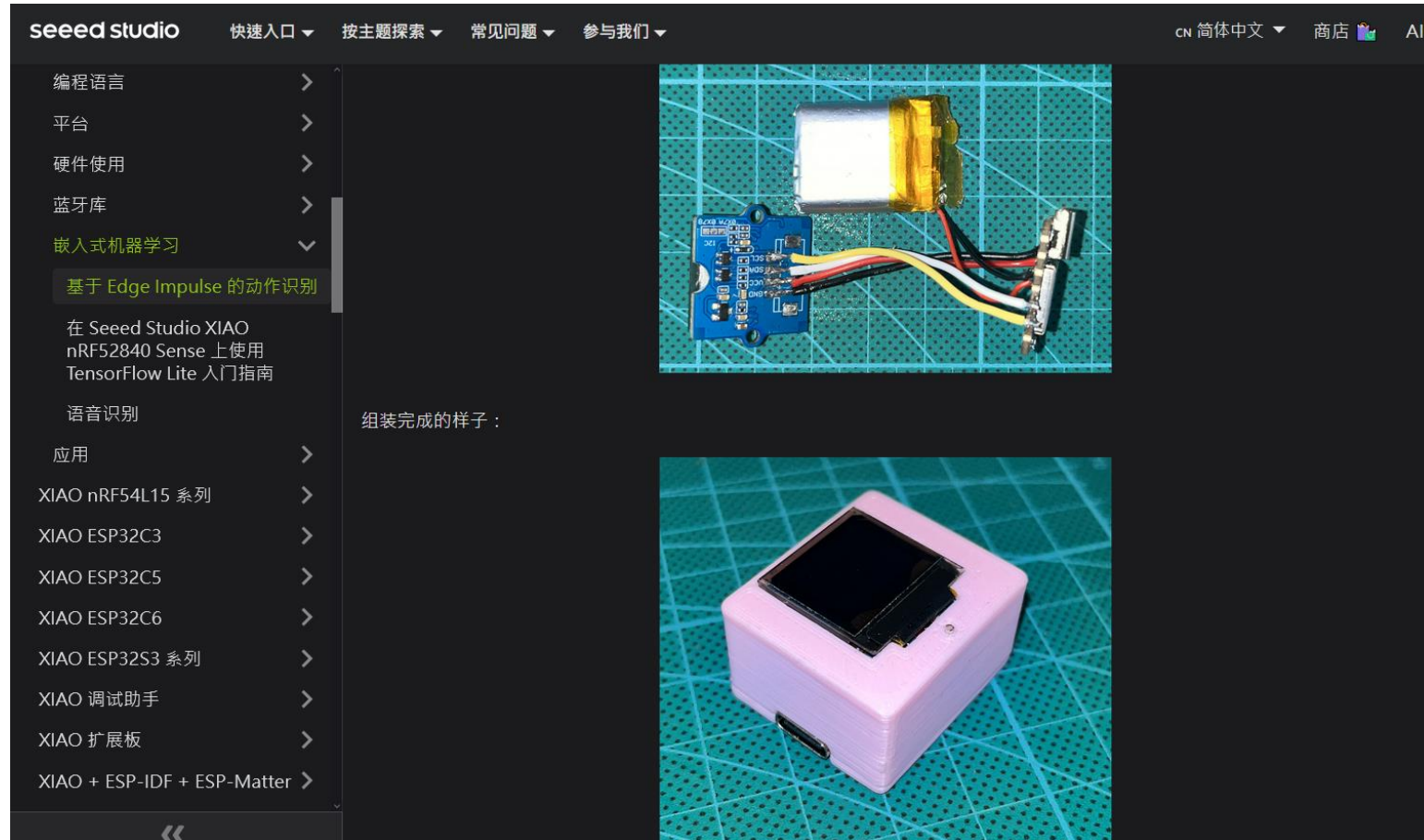
強烈建議一定要採用**Arduino CLI** 增量編譯方式更新程式修改，否則每次將耗費大量時間，增加碳排量。



工作內容	Arduino IDE	Arduino CLI
第一次編譯上傳	19~20min	19~20min
未修改程式第二次編譯上傳	19~20min	< 1min
修改程式後編譯上傳	19~20min	1~2min

<https://omnixri.blogspot.com/2024/10/arduino-20.html>

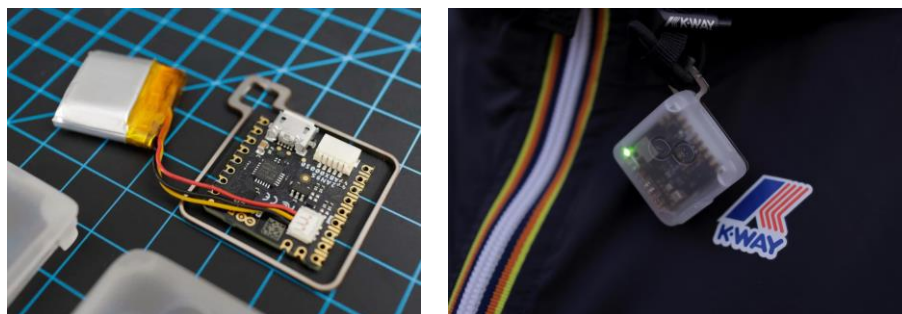
nRF52840 基於Edge Impulse 的動作辨識



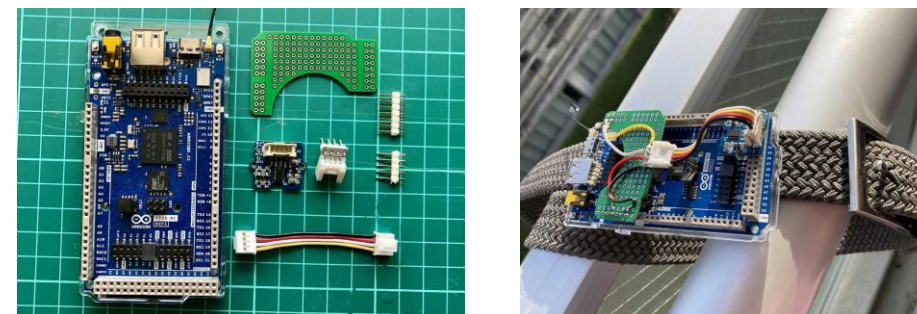
<https://wiki.seeedstudio.com/cn/XIAOEI/>

Edge Impulse Expert 案例分享 (1/2)

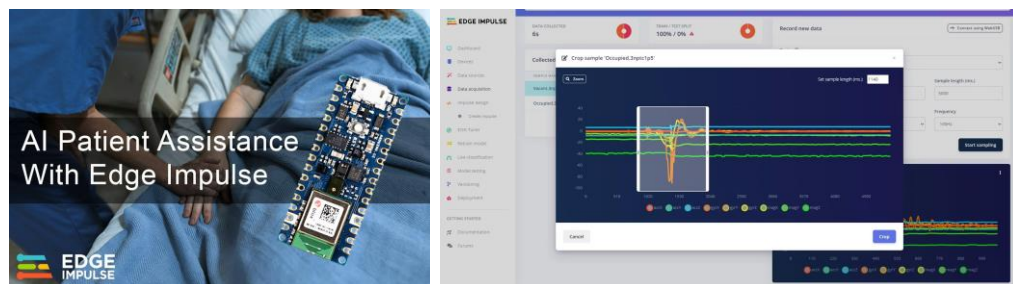
Arduino x K-Way - Outdoor Activity Tracker



Fall Detection using a Transformer Model – Arduino Giga R1 WiFi



Hospital Bed Occupancy Detection - Arduino Nano 33 BLE Sense



Bluetooth Fall Detection - Arduino Nano 33 BLE Sense



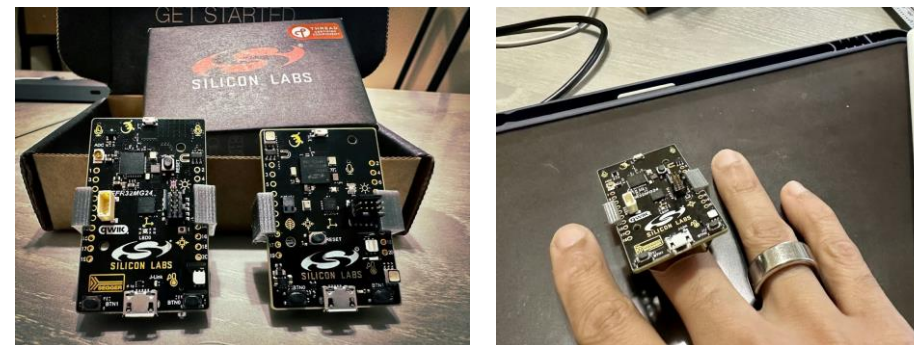
<https://docs.edgeimpulse.com/projects/expert-network/project-list#accelerometer-and-activity-projects>

Edge Impulse Expert 案例分享 (2/2)

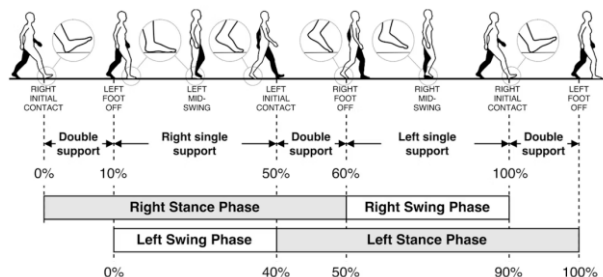
Smart Table Tennis Bat with Live Analytics - Arduino Nano 33 BLE Sense



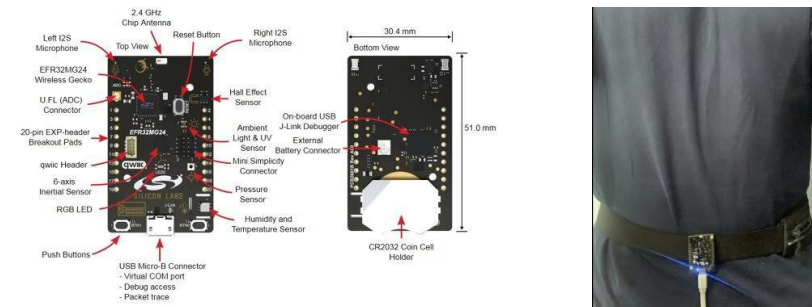
Porting a Gesture Recognition Project from the SiLabs Thunderboard Sense 2 to xG24



Continuous Gait Monitor (Anomaly Detection) - Nordic Thingy:53



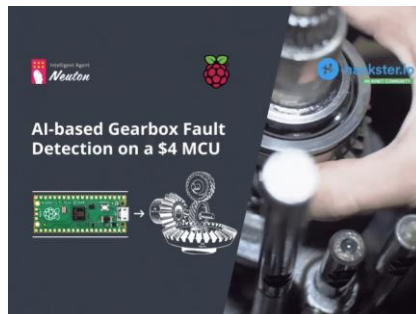
Porting a Posture Detection Project from the SiLabs Thunderboard Sense 2 to xG24



<https://docs.edgeimpulse.com/projects/expert-network/project-list#accelerometer-and-activity-projects>

其它案例分享(1/2)

齒輪箱異常振動偵測



【功能簡述】

利用四個振動感測器產生的資料來分析變速箱是否健康，透過現成資料集進行模型訓練，完成後再部署到開發板上測試，未接真的感測器。

【硬體/週邊】 Raspberry Pi Pico

【軟體/平台】 Neutron Tiny ML Neutron +
Arduino IDE

<https://www.hackster.io/mech/tinymml-gearbox-fault-prediction-on-a-4-mcu-41f434>

包裹運送異常震動偵測



【功能簡述】

利用運動感測器來偵測包裹運送途中的異常狀態，包括亂摔、掉落、搖晃等。

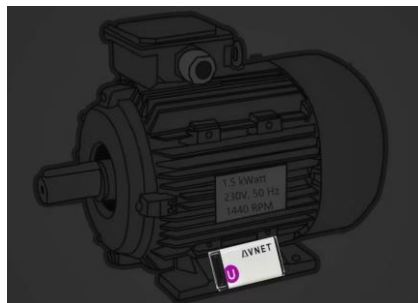
【硬體/週邊】 Arduino Nano 33 BLE Sense +
SIM 800L GSM Module

【軟體/平台】 Edge Impulse Studio +
Arduino IDE

<https://blog.arduino.cc/2022/09/10/this-tinymml-device-monitors-packages-for-damage-while-in-transit/>

其它案例分享(2/2)

馬達異常振動偵測



【功能簡述】

利用開發板上運動感測器產生的資料來分析馬達是否健康，包括

- 軸承故障（例如刺耳的啟動噪音和更大的運轉噪音）
- 對準問題（例如金屬刮擦噪音）
- 機械負載損失（例如不抽水）
- 冷卻風扇損壞（例如風扇葉片在金屬蓋上刮傷）

【硬體/週邊】 Avnet SmartEdge Agile Brainium

【軟體/平台】 Avnet Brainium

https://www.hackster.io/PSoC_Rocks/industrial-motor-diagnostic-866e96

智能桌球拍（虛擬教練）



【功能簡述】

在桌球拍上安裝運動感測器來記錄及分析揮拍姿勢，以利改善後續表現。

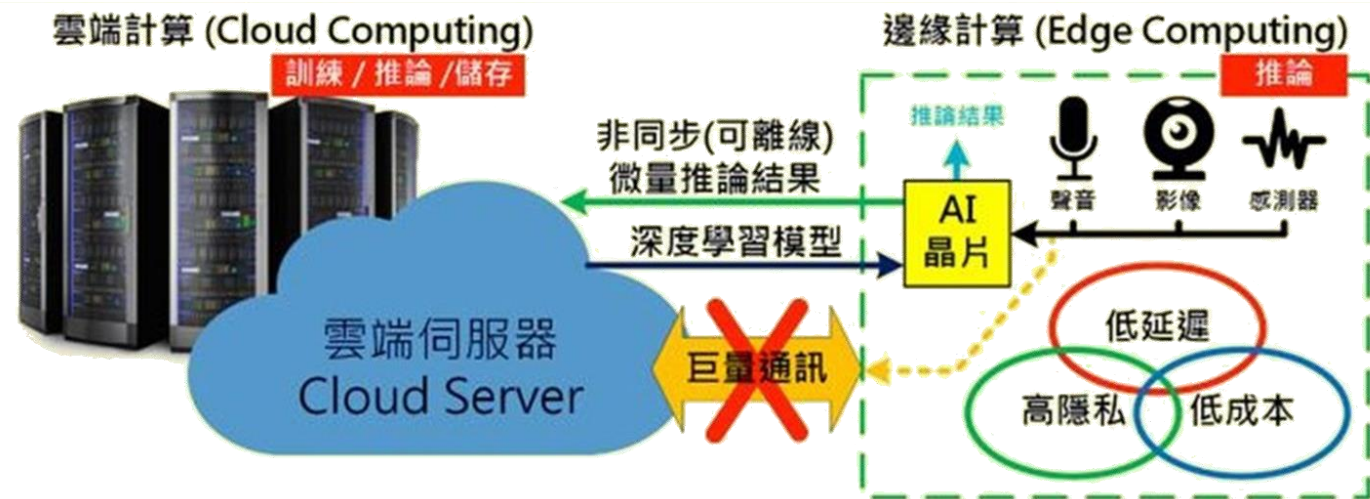
【硬體/週邊】 Arduino Nano 33 BLE Sense + 桌球拍

【軟體/平台】 TensorFlow Lite for Microcontrollers + Google Creativelab - Tiny Motion Trainer + Arduino IDE

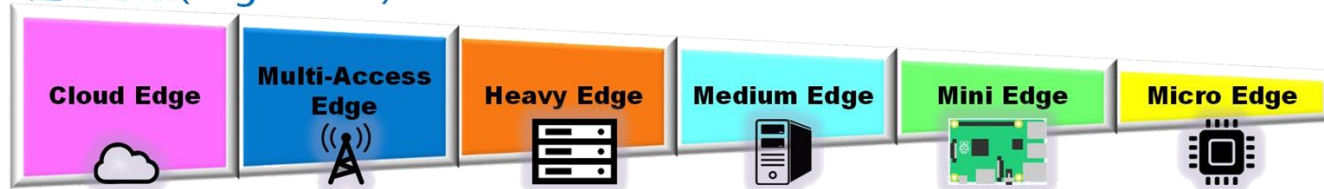
<https://blog.arduino.cc/2022/09/10/this-tinyml-device-monitors-packages-for-damage-while-in-transit/>



Edge AI



邊緣等級(Edge Level)



9.3. 異常振動偵測

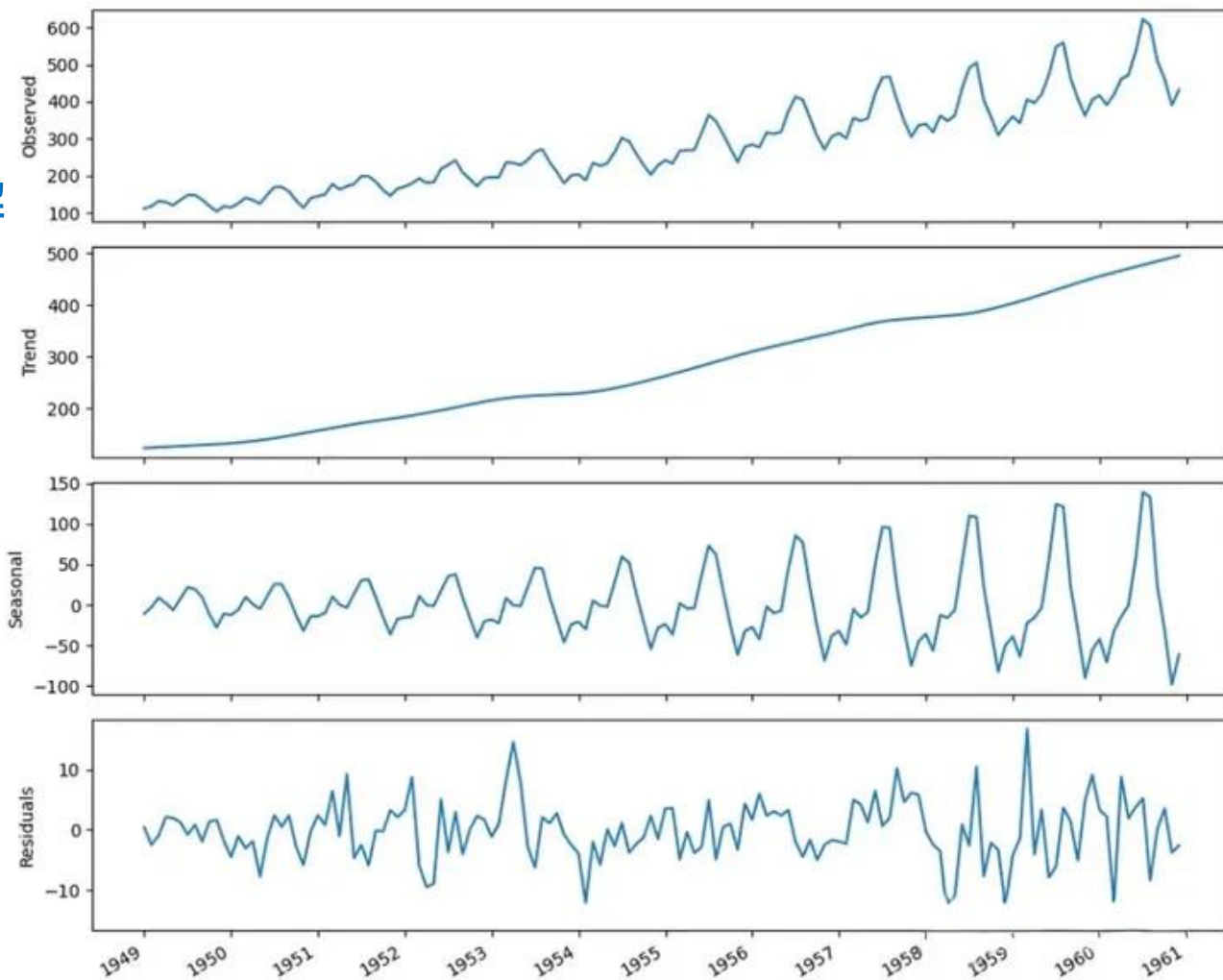
時間序列信號預測

Time Series
原始信號

↓ 分解
趨勢

季節性

殘差



資料來源：<https://vocus.cc/article/6927b46cfd897800014271ee>

➤ 傳統時間序列預測模型

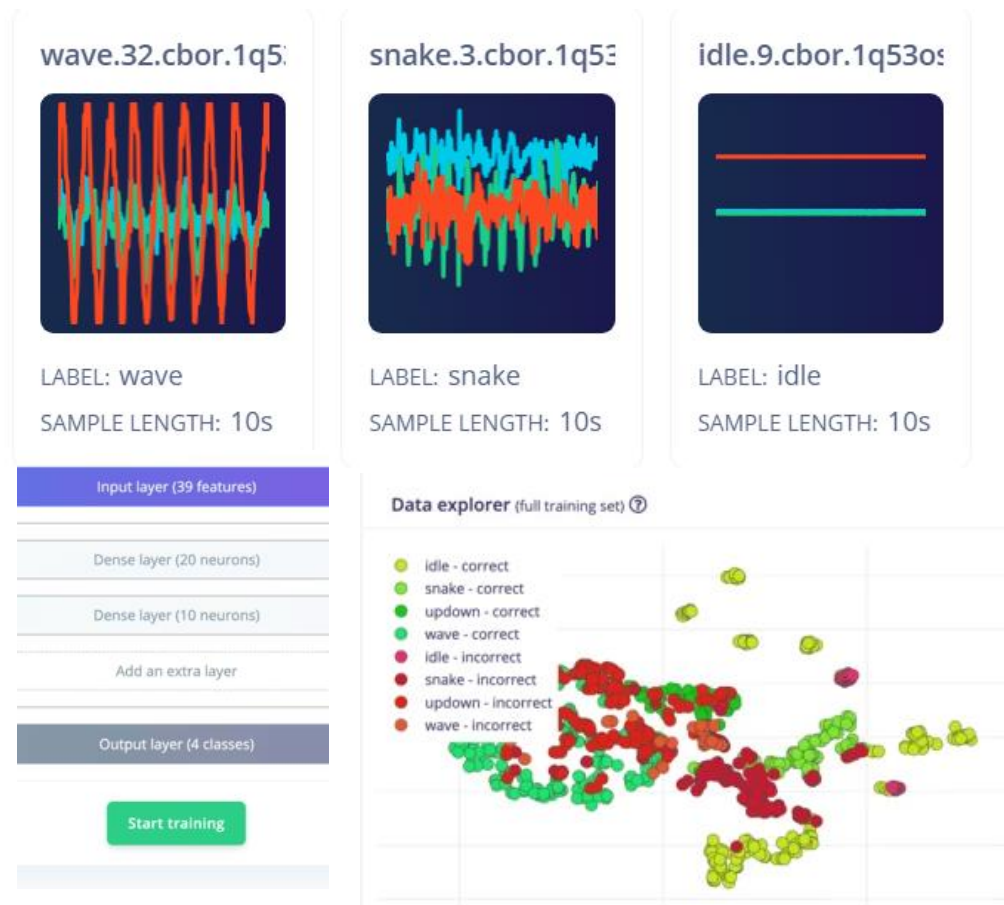
- 自迴歸(AR)
- 移動平均(MA)
- 自迴歸移動平均(ARMA)
- 自迴歸綜合移動平均(ARIMA)
- 季節自迴歸綜合移動平均(SARIMA)
- 帶有外源迴歸量的季節自迴歸綜合移動平均(SARIMAX)
- 向量自迴歸(VAR)
- 向量誤差校正(VECM)

➤ 深度學習時間序列預測模型

- 多層感知器(MLP)
- 迴圈神經網路(RNN)
- 長短期記憶網路(LSTM)
- 自迴歸 LSTMs

常見時序信號異常偵測方式

➤ 特徵樣版分類



➤ 聚類離群偵測



資料來源：<https://docs.edgeimpulse.com/tutorials/end-to-end/motion-recognition>

Edge Impulse 專案 – Impulse Design

EDGE IMPULSE

- Dashboard
- Data acquisition
- Experiments
- EON Tuner
- Impulse design
 - Create impulse
 - Spectral features
 - Classifier
 - Anomaly detection
 - Retrain model
 - Live classification
 - Model testing
 - Deployment

Time series data

Input axes (3)
accX, accY, accZ

Window size
2,000 ms.

Window increase (stride)
240 ms.

Frequency (Hz)
62.5

Zero-pad data

Train on data subset
100 %

Spectral Analysis

Name
Spectral features

Input axes (3)
☒ accX
☒ accY
☒ accZ

Classification

Name
Classifier

Input features
☒ Spectral features

Output features
4 (idle, snake, updown, wave)

Output features

5 (idle, snake, updown, wave, Anomaly score)

Anomaly Detection (K-means)

Name
Anomaly detection

Input features
☒ Spectral features

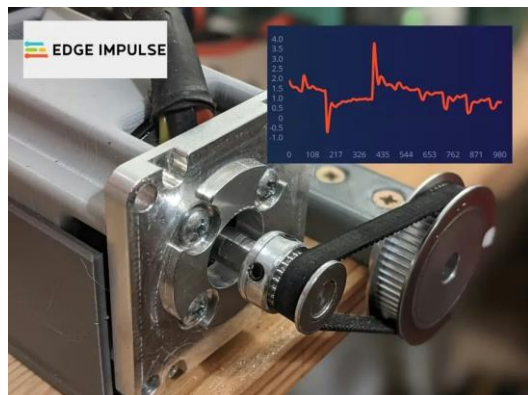
特徵分類

異常偵測

可擇一存在
或同時存在

<https://studio.edgeimpulse.com/public/14299/latest/impulse/1/create-impulse>

練習案例 — 無刷直流馬達異常偵測



Collected data

SAMPLE NAME	LABEL	ADDED	LENGTH
nominal.3292j...	nominal	Yesterday, ...	7m 52s

只收集正常訊號
不需提前分類



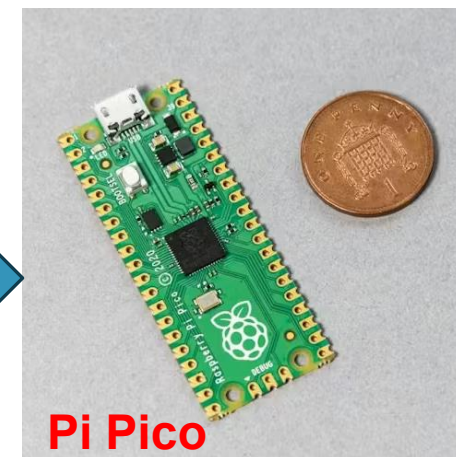
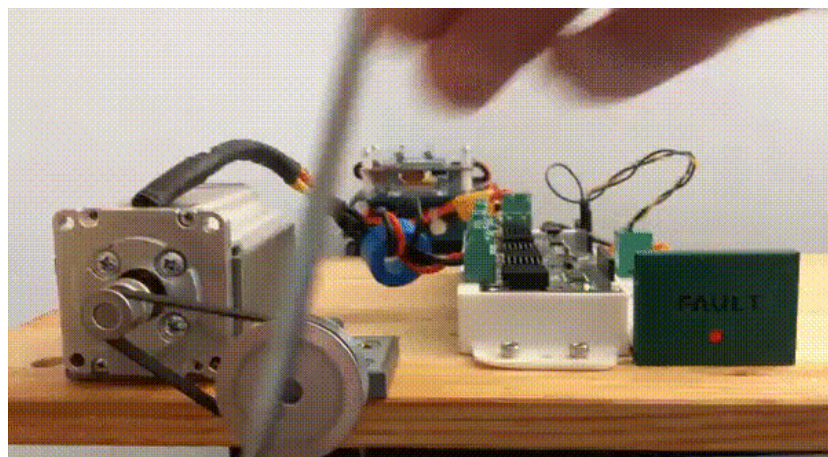
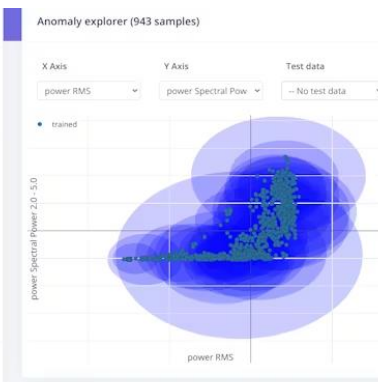
Anomaly detection settings

Cluster count: 32

AXES

- ☒ power RMS
- ☒ power Peak 1 Freq
- ☒ power Peak 1 Height
- ☒ power Peak 2 Freq
- ☒ power Peak 2 Height
- ☒ power Peak 3 Freq
- ☒ power Peak 3 Height
- ☒ power Spectral Power 0.1 - 0.5
- ☒ power Spectral Power 0.5 - 1.0
- ☒ power Spectral Power 1.0 - 2.0
- ☒ power Spectral Power 2.0 - 5.0

Start training



<https://docs.edgeimpulse.com/projects/expert-network/brushless-dc-motor-anomaly-detection>

複製並建立專案

複製專案到個人帳戶

EDGE IMPULSE

Avi / Brushless Motor Anomaly Detection PUBLIC

Target: Raspberry Pi RP2...

Clone this project

Welcome to Edge Impulse, the largest community of edge AI developers!
This is a public Edge Impulse project, use the navigation bar to see all data and models in this project; or clone to retrain or deploy to any edge device.

Brushless Motor Anomaly Detection

About this project

This public Edge Impulse project does not have a README yet. Clone this project to add new data or retrain this project, or to deploy this project to a device.

Data (2 samples)

nominal.3292jcda

LABEL: nominal
SAMPLE LENGTH: 7m 52s

Model accuracy (Unoptimized (float32))

VALIDATION SET TEST SET

Run this model
On any device

Clone project

Dataset summary

DATA COLLECTED
15m 44s

SENSOR
power @ 101Hz

LABELS
nominal

Enter a name for the cloned project:

Brushless Motor Anomaly Detection

Choose your project type:

☒ Personal
60 min job limit, 4GB or 4 hours of data, limited collaboration.

☐ Enterprise
No job or data size limits, higher performance, custom blocks.

Choose your project setting:

☒ Public **選擇公開專案**
Anyone on the internet can view and clone this project under the [3-Clause BSD license](#). Only invited users will be able to edit.

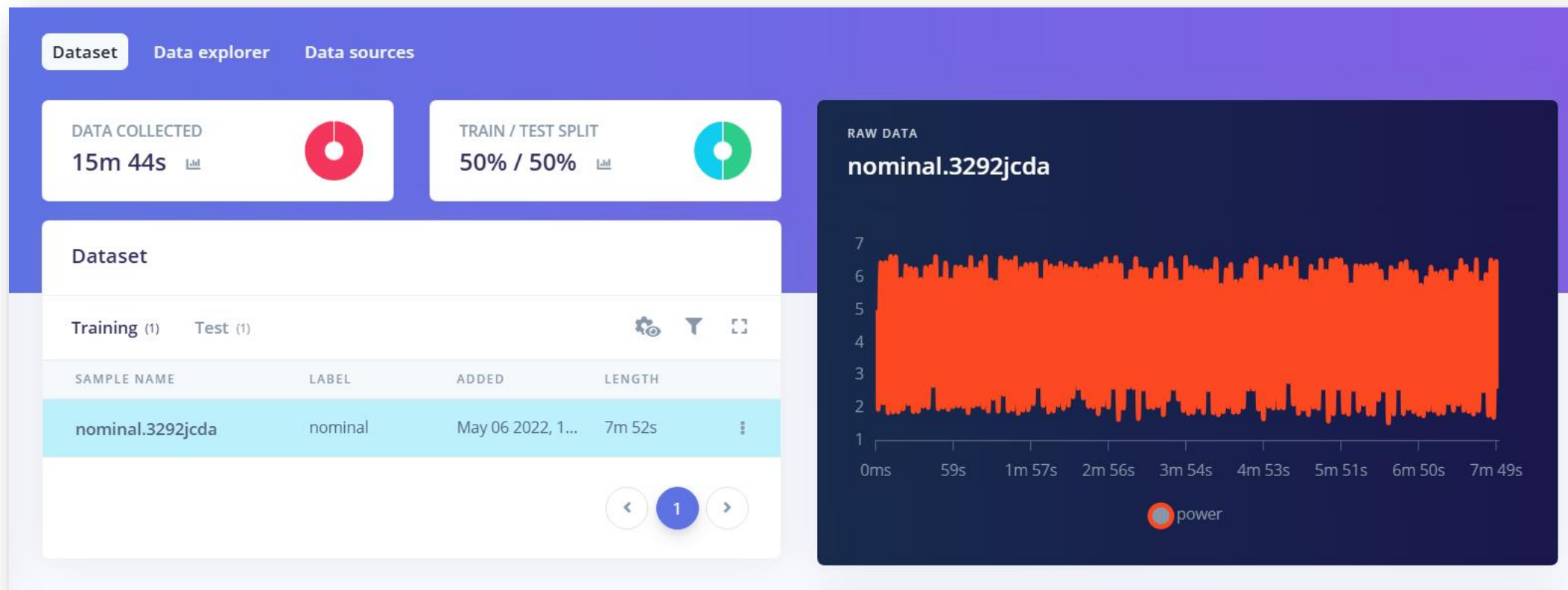
☐ Private (1 of 3 remaining)
Only invited users can edit and view your project.

Clone project

建立專案

<https://studio.edgeimpulse.com/public/102584/latest>

建立資料集



只需收集正常訊號不需提前分類，一筆很長的信號即可，要包含週期變化。

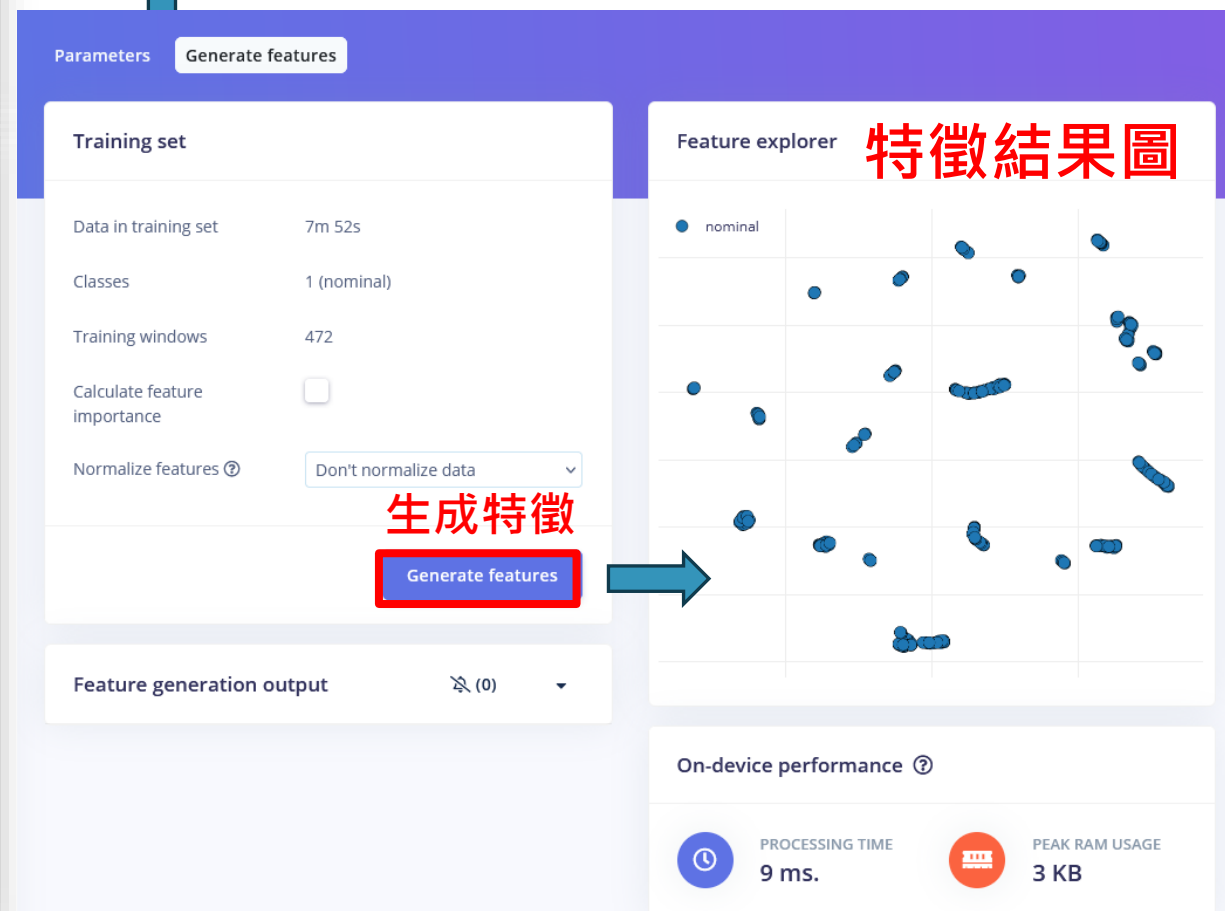
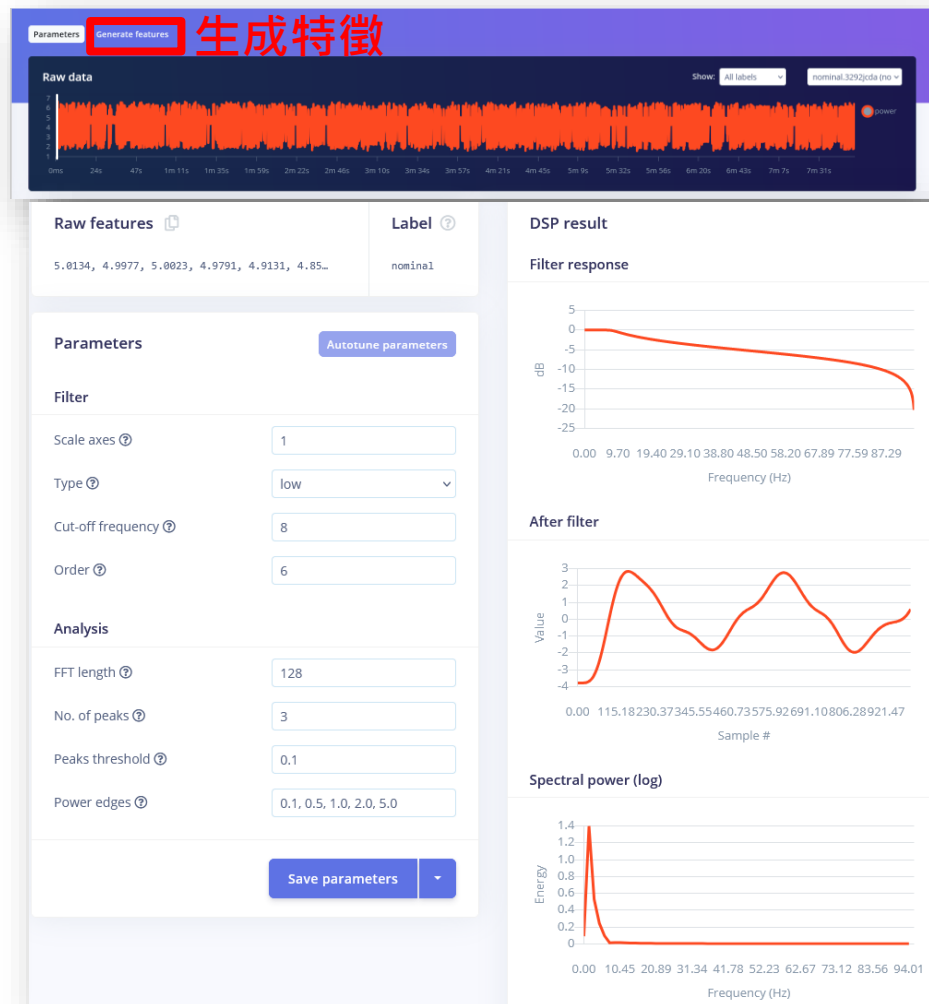
<https://docs.edgeimpulse.com/projects/expert-network/brushless-dc-motor-anomaly-detection>

建立及設定時間序列模型

The screenshot displays the Edge Impulse web interface for configuring a model. The top navigation bar shows the project name 'OmniXRI / Brushless Motor Anomaly Detection' and a 'PERSONAL' label. The sidebar on the left lists various functions: Dashboard, Devices, Data acquisition, Experiments, EON Tuner, Time series data, Create impulse, Spectral features, Anomaly detection, Retrain model, Live classification, Model testing, and Deployment. The main workspace contains four blocks: 'Time series data' (red), 'Spectral Analysis' (white), 'Anomaly Detection (K-means)' (purple), and 'Output features' (teal). The 'Time series data' block has settings for input axes (power), window size (1,000 ms), window increase (stride) (1,000 ms), frequency (191 Hz), zero-pad data (checked), and train on data subset (100%). The 'Spectral Analysis' block has a name 'Spectral features' and input axes (power). The 'Anomaly Detection (K-means)' block has a name 'Anomaly detection', input features (Spectral features), and output features (1 (Anomaly score)). The 'Output features' block has a name '1 (Anomaly score)'. A red box highlights the 'Save Impulse' button, and red text '異常偵測' (Anomaly Detection) and '儲存設定' (Save Settings) are overlaid on the interface.

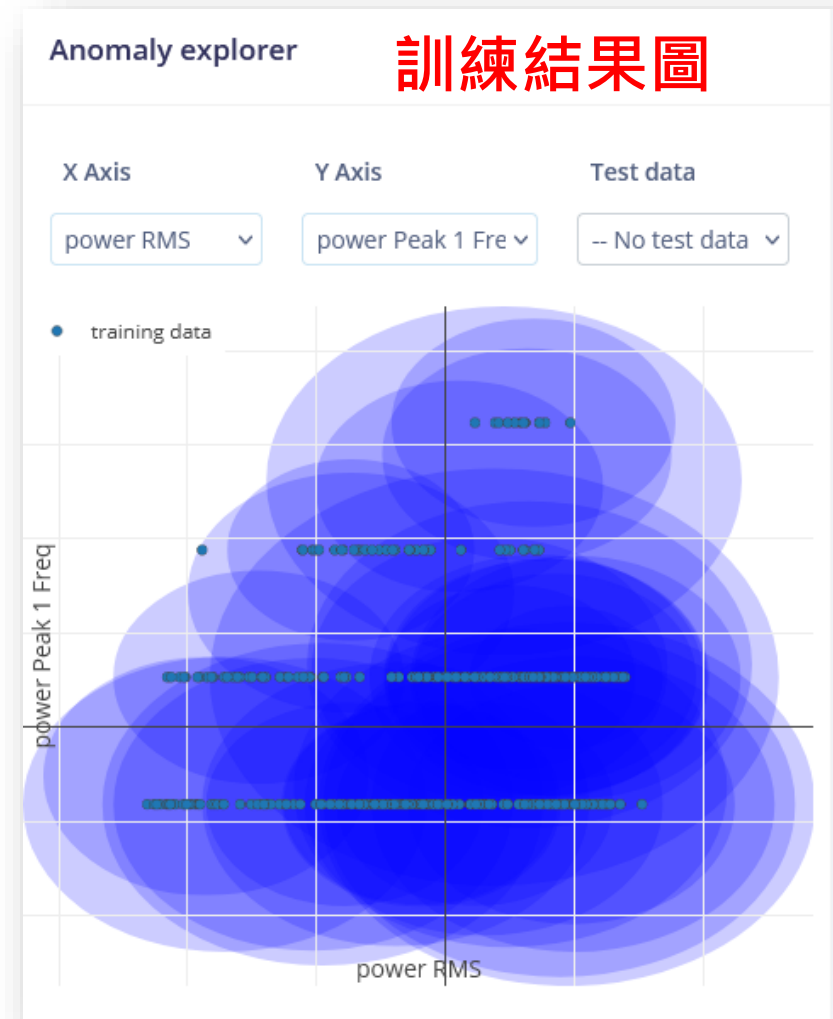
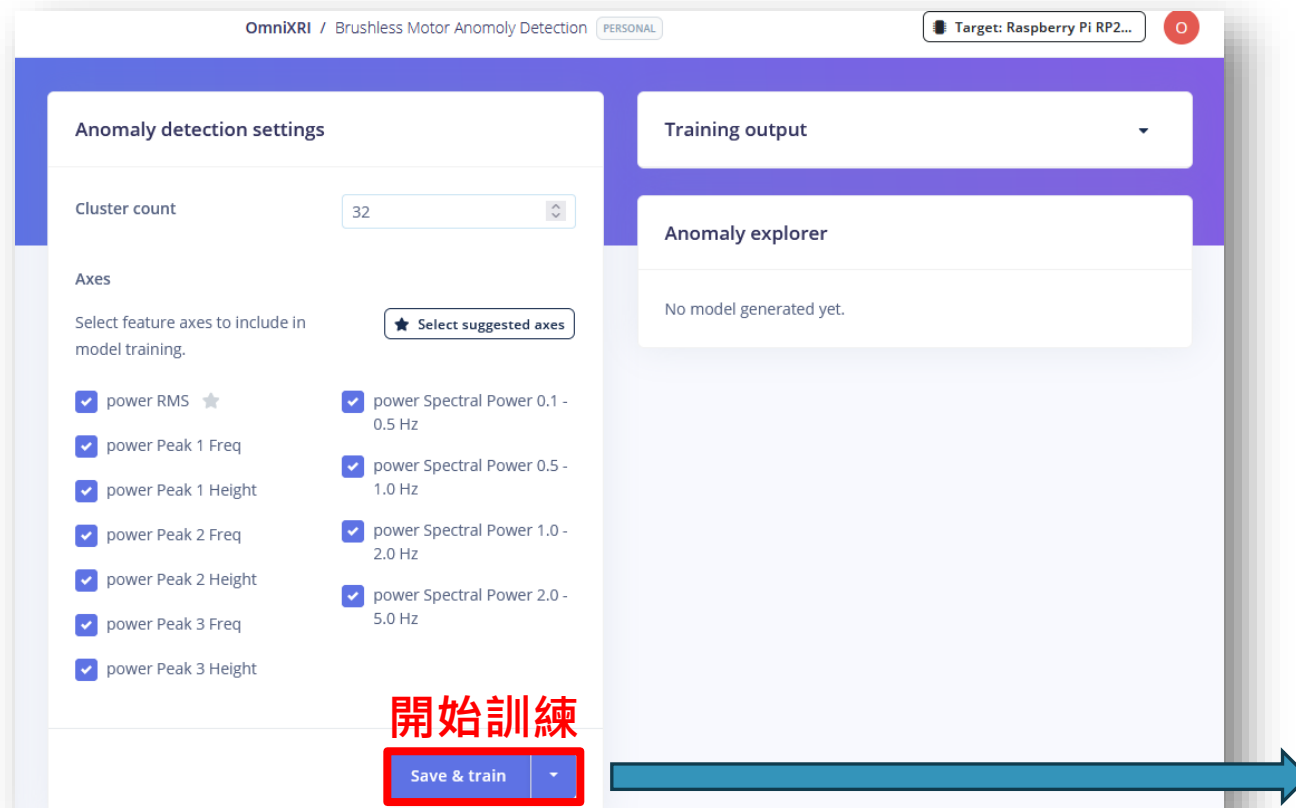
<https://docs.edgeimpulse.com/projects/expert-network/brushless-dc-motor-anomaly-detection>

頻譜分析、特徵生成



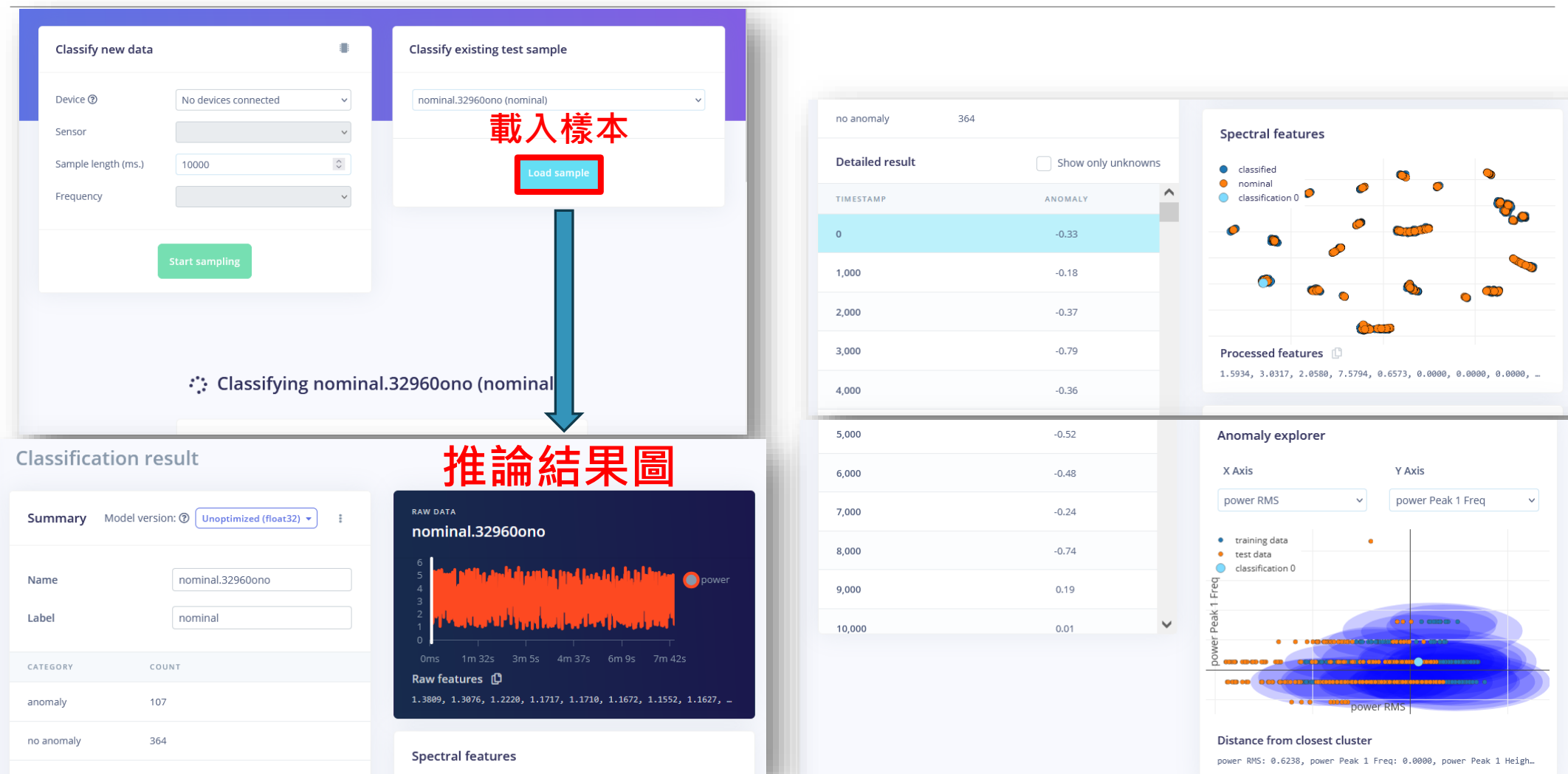
<https://docs.edgeimpulse.com/projects/expert-network/brushless-dc-motor-anomaly-detection>

異常偵測模型訓練



<https://docs.edgeimpulse.com/projects/expert-network/brushless-dc-motor-anomaly-detection>

線上測試



<https://docs.edgeimpulse.com/projects/expert-network/brushless-dc-motor-anomaly-detection>



Arduino 端建立 Pi Pico 開發環境

1. 設定開發板管理員網址

- 開啟 Arduino IDE 2.0。
- 點選選單 [檔案 (File)] > [偏好設定 (Preferences)]。
- 在「其他開發板管理員網址 (Additional Boards Manager URLs)」欄位中，貼上以下連結：
https://github.com/earlephilhower/arduino-pico/releases/download/global/package_rp2040_index.json
(若已有其他網址，請用逗號隔開)
- 按下 [確定 (OK)]。

2. 安裝 Pico 核心

- 點選左側邊欄的 開發板管理員圖示 (或前往 [工具] > [開發板] > [開發板管理員...])。
- 在搜尋框輸入 Pico。
- 找到「**Raspberry Pi Pico/RP2040/RP2350 by Earle F. Philhower, III**」並點選 [安裝 (Install)]。

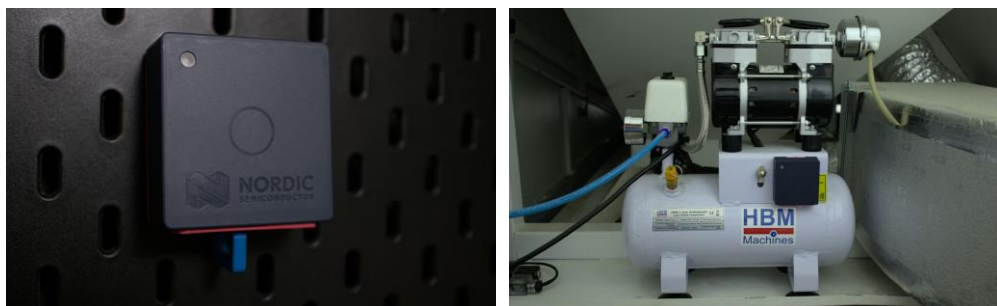
3. 導入 Edge Impulse 生成函式庫

- 開啟選單 Sketch – Include Library – **Add ZIP Library...**
- 加入剛生成的ZIP檔。
- 即可開始程式範例並執行。

<https://docs.edgeimpulse.com/projects/expert-network/brushless-dc-motor-anomaly-detection>

Edge Impulse Expert 案例分享 (1/2)

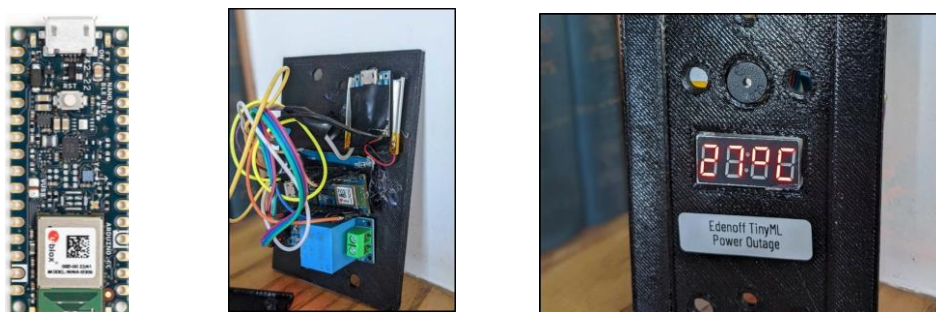
Industrial Compressor Predictive Maintenance - Nordic Thingy:53



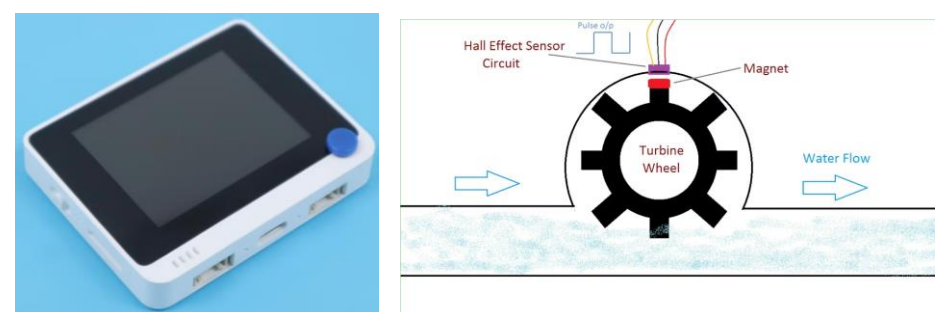
Weight Scale Predictive Maintenance - Arduino Nano 33 BLE Sense



Anticipate Power Outages with Machine Learning - Arduino Nano 33 BLE Sense



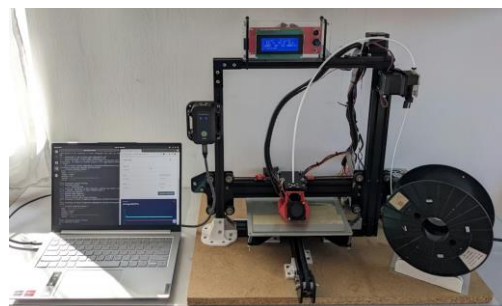
Pipeline Clog Detection with a Flowmeter and AI - Seed Wio Terminal



<https://docs.edgeimpulse.com/projects/expert-network/project-list#predictive-maintenance-and-fault-classification-projects>

Edge Impulse Expert 案例分享 (2/2)

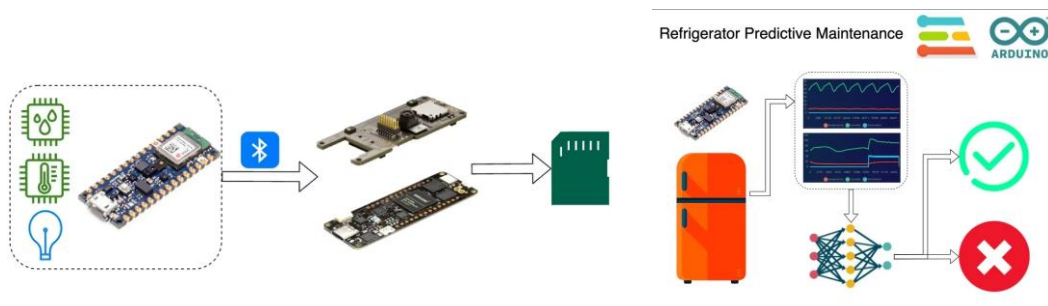
BrickML Demo Project – 3D Printer Anomaly Detection



AI-driven Audio and Thermal HVAC Monitoring – SeeedStudio XIAO ESP32



Refrigerator Predictive Maintenance – Arduino Nano 33 BLE Sense



Motor Pump Predictive Maintenance – Infineon PSoC 6 WiFi-BT Pioneer Kit + CN0549



<https://docs.edgeimpulse.com/projects/expert-network/project-list#predictive-maintenance-and-fault-classification-projects>

參考文獻

- 臺灣科技大學資訊工程系 人工智慧與邊緣運算實務 (CS5149701)
<https://omnixri.blogspot.com/p/ntust-edge-ai.html>
- OmniXRI's Edge AI & TinyML 小學堂 【第13講】 案例實作—運動辨識
https://github.com/OmnixRI/Edge_AI_TinyML_Course_2024/tree/main/Ch13_Motion_Recognition
- Seeed Studio XIAO nRF52840 Sense – 硬件使用 - IMU 使用方法
<https://wiki.seeedstudio.com/cn/XIAO-BLE-Sense-IMU-Usage/>
- Seeed Studio XIAO nRF52840 Sense - 嵌入式機器學習 - 基於 Edge Impulse 的動作辨識
<https://wiki.seeedstudio.com/cn/XIAOEI/>
- 許哲豪，TinyML應用大全 (30組案例分享)
https://hackmd.io/@OmniXRI-Jack/tinyML_30_projects
- Edge Impulse Document – Projects – Expert Network – Accelerometer and Activity Projects
<https://docs.edgeimpulse.com/projects/expert-network/project-list#accelerometer-and-activity-projects>
- Edge Impulse Document – Projects – Expert Network – Predictive Maintenance and Defect Detection Projects
<https://docs.edgeimpulse.com/projects/expert-network/project-list#predictive-maintenance-and-fault-classification-projects>